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BRD4: From molecular understanding to therapeutics development

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SUMMARY

Bromodomain-containing protein 4 (BRD4) is an important therapeutic target for anticancer, antiviral, and anti-inflammatory response. Although its role in transcription, chromatin dynamics, and epigenetic programs is well recognized, recent studies have expanded the function of BRD4 and its related bromodomain and extra-terminal (BET) family members (BRD2, BRD3, and BRDT) to other molecular and biological processes. How a universal epigenetic regulator is tailored for context-specific gene/pathway regulation is mechanistically intriguing. In this review, we highlight the importance of protein isoforms and post-translational modifications, particularly phosphorylation, in generating the protein diversity necessary for selective factor recruitment and context-dependent regulation. Three BRD4 protein isoforms have been identified, including universally expressed BRD4-L and BRD4-S(a) (simplified as BRD4-S), representing long and short isoform a, and cell/stress-specific short isoform b, BRD4-S(b). A phospho-switch mechanism controlling the open/closed state of the bromodomain and BRD4 interaction with partner proteins will also be discussed. Mechanistic understanding of BET protein action has led not only to the development of diverse bromodomain-binding compounds currently used in clinical trials but also to the discovery of a new class of small-molecule inhibitors targeting an intrinsically disordered region (IDR) of phospho-BRD4 to alter specific protein-protein interaction (PPI) networks without globally perturbing transcription programs and chromatin landscapes, thus significantly reducing off-target effects and providing new directions for therapeutic drug development.

In Brief

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Bromodomain-containing protein 4 (BRD4) is an important therapeutic target for anticancer, antiviral, and anti-inflammatory response. In this review, we discuss how BRD4 as a universal epigenetic regulator is tailored for context-specific gene regulation and highlight the importance of protein isoforms and post-translational modifications, particularly phosphorylation, in generating protein diversity and further developing phospho-BRD4 IDR-targeting compounds as new therapeutic agents.

Keywords

BRD4; BET; phosphorylation; PTM; IDR; protein isoform; PPI inhibitor

INTRODUCTION

Bromodomain-containing protein 4 (BRD4) is a member of the bromodomain and extra-terminal (BET) family that also includes mammalian BRD2, BRD3, and the testes/germ cell-specific BRDT. It is evolutionarily conserved in diverse species including Fs(1)h in *Drosophila*, Bdf1 and Bdf2 in *Saccharomyces cerevisiae*,¹ BET-1 in *Caenorhabditis elegans*,² and Brd4 in zebrafish.³ Bromodomains (BDs), comprising ~110 amino acid residues forming four α -helices (α_Z , α_A , α_B , and α_C) and two helix-connecting (ZA and BC) loops, are acetyl-lysine (Kac)-binding modules that typically recognize Kac in both histone and nonhistone proteins.^{1,4} In humans, 61 BDs have been identified and are present in 46 proteins that can be functionally classified into epigenetic regulators, transcription factors, chromatin remodelers, histone-modifying enzymes, tripartite motif (TRIM)-containing proteins, and components of nuclear compartments, among others.^{5,6}

Since the reporting of two Kac-mimicking compounds, JQ1 and I-BET/GSK525762A, capable of reversing cancer-associated phenotypes⁷ and inflammatory responses⁸ by blocking BET bromodomain (BBD) binding to acetylated lysines in histones, hundreds of BBD inhibitors (BBD*i*), broadly or selectively binding one or both of the tandem bromodomains (BD1 and BD2) situated at the N-terminus of each BET protein, have been developed.^{4,9} Many of them are now in clinical trials for treating hematopoietic/solid cancers and cardiovascular/inflammation-associated diseases.⁴ Besides BBD*i*, many protein kinase inhibitors used in patient therapies also show off-target binding to BBDs, partly explaining their clinical efficacy and unanticipated side effects.¹⁰ The pharmacological use of BBD*i*, in conjunction with (or independently of) genetic perturbation of BRD4 expression, in various physiological/pathological settings further reveals BRD4 function in diverse cellular processes (Figure 1), including transcription,^{1,11–20} mRNA processing,^{21–24} DNA replication,^{25–33} DNA damage response (DDR)/repair,^{6,33–40} chromatin organization/maintenance,^{34,41–48} cell cycle progression and checkpoint control,^{49–56} stem cell renewal and lineage differentiation,^{57–66} and viral latency/reactivation,^{13,67–78} as well as cancer development and therapeutics/drug resistance.^{54,55,79–94}

Although BRD4 is a universal epigenetic and transcriptional regulator acting primarily through association with acetylated chromatin and transcription factors at enhancers and promoters, its functional effects on lineage- and cell type-specific transcription programs

are surprisingly selective, e.g., controlling muscle- or fat-specific gene expression during myogenesis and adipogenesis, respectively.^{95,96} This selectivity in target gene response by a universal epigenetic regulator with broad chromatin-binding activity is in part attributed to the differential usage of distinct protein isoforms and post-translational modifications that in combination generate protein diversity and various entities of BRD4 to modulate select factor recruitment and cell type-specific and context-dependent regulation.

In this review, we will summarize the current understanding of BRD4 protein isoforms, their domain features and gene structures, post-translational modifications (PTMs) and phosphorylation-regulated BRD4 function, and the development of a new class of small-molecule inhibitors targeting an intrinsically disordered region (IDR) of phospho-BRD4 that significantly reduces off-target inhibition and potential side effects in therapeutics treatment.

BRD4 Protein Isoforms and Domain Features

Three human BRD4 protein isoforms have been characterized, including BRD4-L (long; NCBI CCDS12328.1/UniProt O60885-1), BRD4-S(a) (short form a, often simplified as BRD4-S; NCBI CCDS46004.1/UniProt O60885-2), and BRD4-S(b) (short form b; NCBI CCDS82307.1/UniProt O60885-3).^{1,54} In addition, a BRD4-NUT (nuclear protein in testis) fusion protein implicated in NUT midline carcinoma (NMC) has also been described.⁸⁹ These BRD4 proteins are generated either by 3' alternative splicing to produce the three mRNA isoforms or by chromosomal translocation to generate the fusion protein (Figure 2). They all share a common N-terminal region encompassing amino acid (aa) residues 1–719 but differ in their C-terminal extension. The common N-terminal region contains tandem bromodomains (BD1 and BD2) with stronger binding to acetylated lysine, weaker affinity towards propionyl-lysine, and no binding to other bulkier acylated lysine residues such as butyryl-lysine and crotonyl-lysine.^{97,98} However, some non-BET BDs, such as the BRD9 bromodomain and the second bromodomain of TATA-binding protein (TBP)-associated factor 1 (TAF1) in TFIID, can bind butyrylated and/or crotonylated lysine residues.⁹⁹

Although BD1 and BD2 are both sequence- and structure-conserved among BET proteins, with each BD more closely related across family members than between BD1 and BD2 within the same protein,¹ they do not bind equivalently to acetylated histones and acetylated transcription factors (TFs). While the acetylated p65 subunit of NF κ B binds both BD1 and BD2 of BRD4,¹⁰⁰ a preference of BD1 for binding to diacetylated histone-like Kac-X-X-Kac peptides derived from TFs GATA1, MyoD1, and E2F1, and chromatin regulators (CRs) ATRX and TOP2A has been reported.^{101,102} Likewise, preferential association of BRD4 BD2 with acetylated TFs TWIST1¹⁰³ and ISX¹⁰⁴, acetylated CR EED1¹⁰⁵, and acetylated Cyclin T1¹⁰⁶ has also been described. For acetylated histone binding, it appears that BD1 and BD2 each have the intrinsic capacity to bind histone H3 lysine 9 (H3K9) and lysine 14 (H3K14) as well as histone H4 lysines K5, K8, K12, and K16 (H4K5, H4K8, H4K12, and H4K16) with unique acetylation patterns determined in part by site-specific histone acetyltransferases (HATs) and histone deacetylases (HDACs) present in each cell type.¹⁴ Any interaction preference between BD1 or BD2 and acetylated histones or TFs, if observed, likely reflects specific PPI networks assembled at particular gene loci in a spatiotemporal and context-specific manner.

With atomic structures determined by multidimensional NMR spectroscopy revealing contact residues between BRD4 BD2 and a K73/K76-diacetylated TWIST1 peptide, it is evident that binding selectivity for BD2 over BD1 is due to a key TWIST1 K73ac contact residue in BD2 (H437) that is replaced by a non-conserved residue in BD1 (D144), even though a conserved asparagine (N) residue important for interaction with Kac residues in histones and TFs is present in both BDs (N140 in BD1 and N433 in BD2).^{10,103} This unique feature has been pharmacologically exploited to develop BD2-selective small-molecule inhibitors for cancer therapy⁸⁸ and for treatment of inflammation-associated cardiovascular and neurodegenerative disorders, among other diseases.^{107,108} Likewise, enhanced target gene selectivity has been described for BD1-selective compounds,^{109,110} based on rational drug design strategies that selectively target BD1 and/or BD2.^{4,9,10}

Besides BD1 and BD2, the common N-terminal region also contains three charged clusters that include a lysine/arginine (K/R) basic residue-enriched interaction domain (BID) and two acidic regions enriched in aspartic acid (D) and glutamic acid (E), designated as N-terminal phosphorylation sites (NPS) and C-terminal phosphorylation sites (CPS; Figure 2).¹⁷ Phosphorylation of the NPS by casein kinase II (CK2) activates the chromatin-binding activity of BRD4 by converting BD2 and BD1 from a closed to an open state, as evidenced by increased sensitivity to Glu-C protease digestion, and promotes the recruitment and stabilization of sequence-specific TFs, such as p53, NF κ B, AP-1, and human papillomavirus (HPV)-encoded E2 protein, for binding to select chromatin loci.^{17,28} Moreover, NPS phosphorylation is reported to enhance BRD4-S dimerization that may contribute to its functional properties.¹¹¹ It is conceivable that phosphorylated NPS (phospho-NPS or pNPS)-modulated BD1/BD2 binding to acetylated chromatin may be similarly employed by phospho-CPS to control the accessibility and/or conformation of the extra-terminal (ET) domain that has been previously shown to interact with distinct chromatin regulators.¹¹²

NPS and BID constitute the core of an IDR that facilitates phase separation and the formation of transcriptionally active biomolecular condensates containing RNA polymerase II (Pol II) complexes.¹¹³ Of note, a nuclear localization signal (KGVKRKADTTTP, spanning aa 286–297) situated within motif A,¹¹⁴ a dimerization region (aa 502–547) designated as motif B,^{111,115} and a reported atypical kinase domain, spanning BD2 to BID, capable of phosphorylating the carboxy-terminal domain (CTD) of Pol II at Ser2 and c-Myc oncoprotein at Thr58¹¹⁶ are also present in the common N-terminal region. The *in vivo* functions of these N-terminal domains, and whether the atypical kinase activity of BRD4 is mechanistically distinct from the other CTD kinases and is not caused by an interacting kinase as recently reported for TAF1,¹¹⁷ remain to be defined.

Unlike the common N-terminal region, the C-terminus is unique to each BRD4 protein isoform. BRD4-L, often interpreted as the full-length protein (aa 1–1362), has a proline-rich (sometimes referred to as proline/glutamine- or PQ-rich) IDR, not found in BRD2, BRD3 and BRDT,^{118,119} that mediates the formation of biologically distinguishable condensates which also contain general transcription cofactor Mediator.^{120,121} Whether this proline-rich C-terminal IDR functions distinctly from or analogously to the charge-enriched internal IDR found in the common N-terminal region in assembling functional biomolecular condensates involved in various aspects of transcription and chromatin dynamics remains

to be investigated. Interestingly, deletion of the C-terminal IDR is reported to convert BRD4-L activity from tumor-suppressive to growth-promoting,¹¹⁸ highlighting the importance of IDRs in regulating the diverse functions of BRD4. A HAT domain (aa 1122–1161) reported within this proline-rich IDR is capable of acetylating histone H3 K122, resulting in nucleosome eviction of BRD4-L from chromatin.⁴² The C-terminal motif (CTM) that mediates interaction with positive transcription elongation factor b (P-TEFb)¹²² and HPV E2^{13,123} is also located at the end of the BRD4-L C-terminus.

BRD4-S(a), lacking the extended C-terminal sequence found in BRD4-L and frequently simplified as BRD4-S, has only three unique amino acids – glycine-proline-alanine (GPA, aa 720–722) – derived from its C-terminal exon and is the universally expressed short isoform that can be easily detected by an S(a)-specific antibody (see Figure 2, antibody panel). While BRD4-L and BRD4-S are present in every cell type examined thus far,^{28,33,54,69,83,93,94,113} BRD4-L is generally the dominant form accounting for more than 87% of total BRD4 in the cell, with BRD4-S representing less than 13%.⁵⁴ Quantification of the relative abundance of BRD4-L/S was initially based on the use of an N-terminal pan-BRD4 antibody, calibrated with known amounts of purified recombinant BRD4-L and BRD4-S proteins, and subsequently validated by using isoform-specific antibodies.⁵⁴

In contrast, BRD4-S(b), which contains a unique C-terminal 75-aa region suitable for generating isoform-specific antibodies (see Figure 2), is not detectable in most cells and appears to be cell type-specific and stress-inducible,³⁴ similar to BRDT that is detectable only in the male and female germ cells.¹²⁴ This unique C-terminal extension of BRD4-S(b) contains a Cys-X-X-Cys motif suggestive of a folded, potentially zinc-coordinating domain that may facilitate the transition of the internal IDR from a disordered-to-ordered structure upon contacting BRD4-S(b)-specific interactors. Indeed, ectopically expressed BRD4-S(b) has been shown to interact with the SMC2 and NCAPD3 components of the condensin II complex to further condense and insulate chromatin from ionizing radiation-induced DDR.³⁴ This is consistent with the observation that knockdown of endogenous BRD4-S(b) increases chromatin accessibility to micrococcal nuclease digestion. Whether the charge-enriched internal IDR and the unique C-terminal extension of BRD4-S(b) are functionally involved in condensin II function remains to be investigated.

In addition to pan- and isoform-specific antibodies, two polyclonal antibodies recognizing pNPS at phospho-S484/S488 (p484/488)¹⁷ and phospho-S492/S494¹²⁵ and two monoclonal antibodies – 8H2 for detection of pan-BRD4²⁶ and 1F11 for detection of BRD4-L (see Figure 2) – are available and have greatly facilitated studies of BRD4 isoforms in various biological settings.

BRD4 Genomic Organization and mRNA Transcripts

The human *BRD4* gene, situated on the short arm (p) 13.12 of chromosome 19 (19p13.12) and spanning nucleotides (nt) chr19:15,235,519–15,332,545 based on the Genome Reference Consortium human genome build hg38 (GRCh38),¹²⁶ is transcribed from the reverse strand of the DNA locus (Figure 3A). According to the Ensembl database¹²⁷ where BRD4 transcripts are manually curated by the Human and Vertebrate Analysis and Annotation (HAVANA) group using Cap Analysis of Gene Expression

(CAGE)-based transcription start site mapping (e.g., oligocapped cDNA libraries and rapid amplification of cDNA ends) and RNA-seq data, BRD4-L is translated from two mRNA transcripts initiated at a proximal transcription start site (TSS) at nt 15,280,451 or a distal TSS at nt 15,332,539, with the proximal TSS also used to transcribe the BRD4-S transcript. Unlike BRD4-L and BRD4-S, the BRD4-S(b) transcript is initiated at nt 15,332,545, i.e., six nucleotides upstream of the distal BRD4-L TSS. Thus, three TSSs (one proximal and two distal) are differentially used to generate distinct BRD4 protein isoforms, with BRD4-L and BRD4-S likely regulated by the same proximal promoter that may be universally expressed. Except for the first exon of each mRNA transcript, which encodes a varied 5' untranslated region (UTR) driven by different promoters, all three BRD4 isoform transcripts share common exons 2–11 encoding the same N-terminal region (aa 1–719). However, due to alternative mRNA splicing at the 3' end, the terminal exon (i.e., exon 12) of BRD4-S and BRD4-S(b) differs from the C-terminal exons (12–20) of BRD4-L, with each isoform transcript also containing a unique 3' UTR and 1–3 consensus poly(A) addition signals (Figure 3B).

From analysis of mRNA splicing ratios (TCGA SpliceSeq) and transcript levels (TCGA Breast Invasive Carcinoma Firehose Legacy RNA-seq datasets) of 709 breast cancer patients, we found that BRD4-L mRNA was approximately twofold more abundant than BRD4-S mRNA and ~56-fold higher than BRD4-S(b) mRNA levels.⁵⁴ Nevertheless, at the protein levels analyzed by isoform-specific immunoblotting with samples from nine human breast cancer lines and two immortalized normal mammary epithelial cell lines, BRD4-S is generally less abundant than BRD4-L, ranging typically from 3–13% of total BRD4, with BRD4-S(b) protein undetectable.⁵⁴ Compared to other triple-negative breast cancer (TNBC) cells, MDA-MB-231 cells, with a nuclear volume of ~250.7 μm^3 ,¹²⁸ have higher concentrations of both BRD4-L (~6.42 μM) and BRD4-S (~0.85 μM). The combined amount of BRD4-L (~970,000 molecules per cell) and BRD4-S (~128,000 molecules per cell), totaling ~1.1 $\times 10^6$ molecules per cell in MDA-MB-231 cells,⁵⁴ is ~3.4-fold higher than that of Pol II (~320,000 molecules per cell in HeLa cells)¹²⁹ and substantially exceeds the levels of c-Myc (~60,000 molecules per cell) and p53 (~160,000 molecules per cell), both measured in ER/PR-positive MCF-7 breast cancer cells.¹³⁰ This comparison, albeit with numbers deduced from different human cancer lines, indicates that BRD4 is an abundant transcriptional regulator with additional roles in genomic organization, chromatin dynamics, and other cellular processes (see Figure 1).

The disparity between transcript and protein levels suggests that steady-state levels of BRD4 protein isoforms, besides being controlled by the differential promoter and TSS usage, can also be modulated by post-transcriptional and/or post-translational regulation, such as 5'-capping, pre-mRNA splicing, 3'-end mRNA processing, nucleocytoplasmic transport, mRNA decay, ribosome entry/scanning through distinct 5'UTRs, and protein turnover. This multi-tiered regulation is supported by observations that (1) BRD4-L/S pre-mRNAs harbor multiple AAUAAA polyadenylation signals that can be selectively used to generate alternative 3'UTRs susceptible to microRNA (miRNA)-mediated repression, as demonstrated by reduced BRD4-L protein levels in cardiomyocytes following miR-9 binding to the 3'UTR of BRD4-L mRNA¹³¹ and (2) elevated levels of BRD4 deubiquitinase DUB3 coincide with increased BRD4 protein levels in some prostate cancer cells.¹³²

It is worth mentioning that although the mRNA level of BRD4-S(b), as determined by RNA-seq, is much lower than that of BRD4-L and BRD4-S, the RNA reads aligned to the first exon and intron of nascent (i.e., newly-synthesized) BRD4-S(b) transcripts, as measured by precision run-on sequencing (PRO-seq), are in fact comparable to those mapped to the common exons/introns shared by all three BRD4 isoform transcripts.³³ This finding suggests that a low level of BRD4-S(b) protein, if detectable, may not be solely attributed to transcriptional regulation but instead reflects differences in mRNA processing, modification, decay, or other undefined post-transcriptional processes. A recently described noncoding genetic alteration mechanism involves recurrent focal deletions within the 5'UTR-coding region of the BRD4 gene, spanning the distal TSS exon 1 and intron 1.¹³³ These deletions are frequently observed in breast, ovarian, and endometrial cancers with chromosome 19 amplification and appear to mitigate the potential toxicity associated with overexpression of BRD4-L and/or BRD4-S(a). This finding underscores the importance of gene dosage balance to fine-tune BRD4 isoform expression in sustaining cancer cell growth.

BRD4-Regulated Transcription and Chromatin Targeting

The transcriptional role of BRD4 – typically referring to BRD4-L owing to its cellular abundance when no particular isoform is designated – was first established in the mid-2000s based on findings that BRD4 enhances human immunodeficiency virus (HIV) gene transcription by recruiting the P-TEFb kinase to the HIV promoter^{11,12} and that in the context of HPV, BRD4 stabilizes E2 binding to the HPV E6 promoter-proximal region, thereby preventing TFIID/Pol II promoter association and suppressing E6/E7 oncogene expression.^{13,134} This dual function of BRD4 in gene activation and repression is commonly utilized by both viral and cellular transcription factors and cofactors^{1,135} and is frequently observed in genome-wide transcriptome analysis.^{18,19,33,39,45,54,55,65,66,83,87,91,93,96,105,136–138}

In Pol II-driven early transcription, four BRD4-regulated steps have been defined – commitment, transcription initiation, promoter-proximal pausing, and pause release (Figure 4A):

1. Commitment: Commitment is the key step underlying how a broadly acting Kac-binding chromatin regulator, such as BRD4, achieves gene-specific transcriptional control. This process is primarily determined by BRD4 working in conjunction with transcriptional and chromatin regulators that provide sequence and/or locus specificity through (1) direct interaction with sequence-specific transcription factors (TFs)^{1,17,28,54,136,139} and chromatin regulators (CRs),^{16,69,105,112} which include histone-modifying enzymes and chromatin-remodeling factors, and (2) association with noncoding RNAs (ncRNAs), e.g., enhancer RNAs, that may bind directly to either or both BDs¹⁴⁰ or indirectly with BRD4 through N⁶-methyladenosine (m⁶A)-modified ncRNA reader proteins, such as YTH domain-containing protein 1 (YTHDC1)¹⁴¹ (Figure 4A, 1st panel). Notably, the imprinted ncRNA H19 selectively associates with BRD4, but not BRD2 or BRD3, to control migration-related gene expression,¹⁴² while unique ncRNAs derived from gamma satellite DNA II (GSATII) also use m⁶A-YTHDC1-BRD4 enhancer association to control transcription of pericentromeric and centromeric repetitive elements.¹⁴³

The association of BRD4 with distinct TFs/CRs is mediated by different regions of BRD4, underlying the scaffolding role of BRD4 in promoting functional transcription/chromatin complex assembly (Figure 4B). BD1 and BD2 are acetylated chromatin-binding modules that are also targeted by some ncRNAs likely in a sequence/structure-dependent manner and by select Kac-containing proteins through their acetyl-lysine interaction with the hydrophobic pockets of BD1 and/or BD2 (see above, and the BRD4 Protein Isoforms and Domain Features section). While BD1/BD2-dependent acetylated chromatin binding seems to be the major determinant for genome-wide BRD4 association, acetylation-independent BRD4 recruitment to gene targets may also occur through direct interaction with TFs, CRs, Pol II machinery, enhancer/promoter-associated ncRNA, or DNA^{113,144,145} in nucleosomes or nucleosome-free regions.

Besides BD1/BD2-mediated protein association, phosphorylation of NPS provides a separate platform for BD-independent interaction with distinct pNPS reader proteins, such as DDR factors 53BP1, BARD1, and other proteins containing phosphoserine/threonine (pS/T)-binding modules that include breast cancer gene 1 (BRCA1) C-terminal (BRCT) domain, Fork-head-associated (FHA) domain, polo-box domain (PBD), FF domain, and 14-3-3 family proteins.³³ Additionally, pS/T reader-independent interaction has also been found in other pNPS binders such as p53,¹⁷ both p65 and p50 subunits of NFκB, the FosB member of dimeric AP-1 complexes, high-risk (i.e., cancer-associated) HPV-encoded E2 proteins,²⁸ and BRD4-interacting chromatin-remodeling complex-associated protein (BICRA, also named GLTSCR1), which is uniquely found in the non-canonical SWI/SNF complex.¹⁴⁶ Other BRD4-interacting proteins binding to the ET domain (e.g., ATAD5, CHD4, JMJD6, NSD3, KSHV-encoded latency-associated nuclear antigen LANA-1, MLV-encoded viral integrase, cohesin-loading factor NIPBL) or the CTM (e.g., P-TEFb, histone H3K4 lysine demethylase KDM5C, and E2 encoded by all HPV types) have been discussed.^{29,45,112,136,147–150} Clearly, association with distinct TFs, CRs, and ncRNAs through BD1/BD2-dependent and BD-independent pNPS/ET/CTM and presumably BID binding creates diverse BRD4-regulatory platforms for the *commitment* step of gene-specific targeting.

It should be mentioned that commitment initiated by transient binary association at an enhancer or promoter region can either be reversed by negative regulators or gradually stabilized by positive factors through sequential entry of additional proteins or macromolecules to build up a dedicated transcription complex at a specific locus to a point of no return. The role of BRD4-L and BRD4-S in gene commitment can be independent, synergistic, or antagonistic in part determined by the specific gene locus and the biological context in an often spatial and temporal dependent manner. The expression of extracellular matrix (ECM) network (i.e., matrisome) genes opposingly controlled by BRD4-S and BRD4-L through their respective association with enhancers and promoters to activate or repress matrisome network gene transcription elegantly illustrates the dynamic regulation at the commitment step.⁵⁴

2. Transcription initiation: Once chromatin targets are engaged, BRD4 interacts with general cofactor Mediator, which in turn recruits Pol II, along with its auxiliary general transcription factors (GTFs) including TFIIA, TFIIB, TFIIE, TFIIF, and TFIIH, to TFIID-bound promoters through either a stepwise preinitiation complex (PIC) assembly pathway

or a preassembled Pol II holoenzyme pathway.^{135,151,152} Pol II then initiates transcription from class II promoters (Figure 4A, 2nd panel) upon phosphorylation by the Cyclin H/CDK7 kinase module of TFIIF at the serine 5 (Ser5) residue of the heptapeptide repeats (YSPTSPS) in the CTD of its largest subunit, RPB1.

Association between BRD4 and Mediator was first identified by anti-FLAG antibody-mediated immunoprecipitation (IP)-coupled Western blotting (WB) and mass spectrometry (MS) analysis of FLAG-tagged BRD4-L complexes from cellular extracts,^{11,12} and subsequently by identification of BRD4-L in purified FLAG-tagged human Mediator complexes with or without the Cyclin C/CDK8 module.^{1,153} It appears that Cyclin C/CDK8 in the kinase module is not essential for Mediator interaction with BRD4-L. With IP-proteomic analysis of endogenous BRD4-L and BRD4-S complexes,³³ specific Mediator subunits present in the BRD4-L complexes include MED13/CDK8 in the kinase module, MED6/8/11/17/18/22/27/28/30 in the head module, MED1/4/7/9/10/21/26/31 in the middle module, and none in the tail module. In contrast, only a few Mediator components, such as kinase module subunits MED13/13L/CDK8 and middle module subunit MED1, are identified in the BRD4-S complexes.³³ This differential association of Mediator with BRD4-L versus BRD4-S, initially observed in HPV-harboring keratinocytes,³³ is also seen in HeLa cervical cancer and MDA-MB-231 breast cancer cells, likely reflecting a role of the unique C-terminal IDR of BRD4-L in facilitating compartmentalization and enrichment of MED1-IDR within specific biomolecular condensates.¹²⁰ This BRD4-L-Mediator interaction appears to increase transcriptional output through association with super-enhancers in cancer cells.¹²⁰ Interestingly, BRD4-S can also form biomolecular condensates in C-33A cervical cancer cells, correlating with enhanced Pol II transcription and BRD4-L/MED1 association.¹¹³ It would be of great interest to examine whether the composition of BRD4-S condensates is similar to or distinct from those formed by BRD4-L.

The requirement of BRD4 and Mediator for transcription initiation, however, is likely chromatin-dependent, since both basal and activator-dependent transcription from nucleosome-free class II genes can readily occur in highly purified *in vitro*-reconstituted transcription systems driven by TBP-initiated stepwise recruitment of TFIIB, Pol II/TFIIF, TFIIE, and TFIIF to TATA box-containing DNA templates, independently of TBP-associated factors (TAFs) present in TFIID and in the absence of BRD4 and Mediator.^{153–159} Thus, the action of BRD4 and Mediator may be to facilitate Pol II and GTF, especially TFIID, access to chromatin-restricted promoter/enhancer regions in *in vivo* settings. This is consistent with findings that (1) both BRD4 and Mediator are critical for forming transcriptionally active biomolecular condensates;^{113,120} (2) Bdf1, the yeast homolog of mammalian BRD4, is highly enriched at the promoter-proximal +1 nucleosome situated immediately downstream of the TSS;¹⁶⁰ (3) Bdf1 has sequence homology to the C-terminal part of human (h) TAF1 in TFIID and interacts with the yeast homolog of hTAF7;¹⁶¹ and (4) recombinant hTAF7 interacts through its central region with hTAF1 and its N-terminal region with many transcriptional activators including Sp1, YY1, USF, CTF, adenoviral E1A, and HIV Tat proteins,¹⁶² and binds BRD4-L in a region spanning BD2-NPS-BID.¹¹⁶

The presence of tandem BDs in hTAF1 for binding to diacetylated histone H4¹⁶³ and a unique plant homeodomain (PHD) finger in hTAF3 for contacting the trimethylated

histone H3 lysine 4 (H3K4me3) promoter mark in active genes¹⁶⁴ further highlights the key role played by TFIID as a core promoter-binding factor, particularly in the context of chromatin.¹⁵⁸ Although TFIID and Mediator may be functionally equivalent in some gene contexts,¹⁵³ how BRD4 functions through TFIID to regulate Pol II-dependent transcription in mammalian cells is less clear and would be of great interest for future studies.

Besides short-range proximity-driven gene control, BRD4 also mediates higher-order chromatin organization in oncogene-amplified extrachromosomal DNA (ecDNA) formation⁴⁶ and ultralong-range enhancer-enhancer contacts.⁴⁷ It seems that transcription of the ecDNA-amplified oncogene (e.g., *c-Myc*) and its adjacent long ncRNA (e.g., *PVT1* for plasmacytoma variant translocation 1) gene loci is critical for maintaining ecDNA cluster number/size, nuclear compartmentalization in interphase, and mitotic segregation.^{165,166} Due to the lack of centromeres needed for faithful mitotic segregation, ecDNA may anchor to mitotic chromosomes via BRD4 association, analogous to that observed in BRD4-facilitated extrachromosomal HPV (ecHPV), Epstein-Barr virus (ecEBV), and Kaposi's sarcoma-associated herpesvirus (ecKSHV) genome segregation in mitosis.^{167–169} Whether active prophase ecDNA transcription or a simple physical (i.e., transcription-independent) BRD4-ecDNA association is essential for stable ecDNA inheritance through mitosis, with an unequal segregation of ecDNA copies in daughter cells that partly accounts for intratumoral heterogeneity, remains to be elucidated. Perhaps live-cell imaging directly monitoring active vs. paused Pol II association or enhancer/enhancer vs. enhancer/promoter contact changes, with or without genetic/chemical perturbation of BRD4 and/or histone acetylation,²⁰ will help distinguish the roles of transcription vs. chromatin dynamics in ecDNA inheritance. Clearly, the mechanisms underlying BRD4-L/S-modulated ecDNA formation and long/ultralong-range enhancer-enhancer networks in gene commitment and TFIID/Mediator-mediated Pol II recruitment remain to be further defined.

3. Promoter-proximal pausing: Following initiation of transcription, the initial synthesis of nascent RNA phosphodiester bonds by Ser5-phosphorylated Pol II often pauses at a promoter-proximal region around 50 nt or so (Figure 4A, 3rd panel). This promoter-proximal pausing occurs in over 90% of Pol II-bound mammalian genes^{16,170} and is typically enhanced by the presence of DRB (5,6-dichloro-1-β-D-ribofuranosylbenzimidazole) sensitivity-inducing factor (DSIF; heterodimer of SPT4 and SPT5) together with negative elongation factor NELF containing four subunits (NELF-A, NELF-B, NELF-C/D, and NELF-E).^{171–175} However, in a nucleosome-free *in vitro*-reconstituted transcription system, promoter-proximal pausing seems to be TAF-dependent, since only the TFIID- but not TBP-assembled PIC can recapitulate promoter-proximal pausing,¹⁵⁹ which is further supported by the identification of a consensus TFIID-binding signature deduced from over 177,000 human core promoters likely to be critical for TFIID-mediated Pol II promoter-proximal pausing.¹⁷⁶ The coincidence of the BRD4-enriched +1 nucleosome with acetylated histones immediately downstream of paused Pol II (Figure 4A, 3rd panel) suggests that BRD4 and the +1 nucleosome may also play a key role in Pol II pausing.

4. Pause release: While the involvement of BRD4 in Pol II pausing remains to be defined, it is well-documented that BRD4 controls the efficiency of Pol II release from the promoter-proximal pause sites (Figure 4A, 4th panel). Promoter-proximal pausing serves as a platform for factor exchanges, particularly the dissociation of initiation factors and entry of elongation factors, 5'-capping enzymes, mRNA splicing factors, and 3'-mRNA processing and termination factors. The importance of BRD4 as a 5' elongation checkpoint factor mediating Pol II pause release is elegantly illustrated by defects in pause release, 3'-end mRNA processing, and transcription termination upon depletion of endogenous BRD4-L and BRD4-S, but not BRD2 and BRD3, using genome editing and a BRD4-specific degrader.²³ It is clear that pause release requires BRD4 association with key elongation factors such as DSIF, Pol II-associated factor 1 complex (PAF1c), both Cyclin T1 and CDK9 subunits of P-TEFb, and histone chaperone/elongation factor SPT6,^{17,19,23} while the presence of 3'-end mRNA cleavage and polyadenylation specificity factor (CPSF) and cleavage stimulatory factor (CstF) in the BRD4 complex is critical for 3'-processing and transcription termination.²³

Mechanistically, BRD4-recruited P-TEFb kinase phosphorylates the Pol II CTD at Ser2, DSIF subunit SPT5, and NELF subunits A and E, resulting in NELF dissociation, PAF1c binding, and release of paused Pol II.^{177,178} While BRD4 is essential for P-TEFb recruitment, its bromodomains seem dispensable since deletion of BD1 and BD2 does not affect BRD4 and P-TEFb association nor Pol II pause release.¹⁷⁹ Such BD-independent interaction has also been observed in BD-deleted BRD4 association with Mediator, SPT5/DSIF, and SPT6,¹⁸⁰ thus providing a molecular switch mechanism underlying BRD4 recruitment from bromodomain-dependent chromatin binding during initial breast cancer development to bromodomain-dispensable but Mediator-dependent association when tumors become insensitive to BET inhibitor treatment.⁸³

Depending on the molecular features of Pol II stalling, additional mechanisms have been described for release of promoter-proximal pausing. P-TEFb, besides existing as an active Cyclin T1-CDK9 dimer, is also sequestered as an inactive small nuclear ribonucleoprotein (snRNP) complex containing 7SK small nuclear RNA (snRNA) and HEXIM1/2, MEPCE, and LARP7 proteins.¹⁸¹ Release of this inhibitory complex from the pause site can be potentially achieved by BRD4-dependent recruitment of JMJD6 (Jumonji domain-containing protein 6) to the anti-pause enhancer (A-PE).¹⁶ The A-PE-associated JMJD6 acts to remove the repressive dimethylated histone H4 arginine 3 (H4R3me2) chromatin mark and the 5'-methyl-cap in 7SK snRNA through long-range A-PE contact with the paused promoter bridged by BRD4, leading to resumed elongation of Pol II. Other pause-release mechanisms may also occur via BRD4-mediated recruitment of TFs or DNA/chromatin regulators to the pause sites. Documented examples include BRD4-recruited c-Myc¹⁷ to facilitate Pol II pause release rather than Pol II recruitment¹⁷⁰ and BRD4-mediated Pol II CTD Ser2 phosphorylation that in turn stimulates DNA topoisomerase I activity to help overcome transcription-induced torsional stress in DNA topology after pause release.¹⁸² Moreover, how the positioning of the BRD4-enriched +1 nucleosome,¹⁶⁰ promoter-proximal DNA/RNA sequence and structure,^{176,183} and m⁶A-modified nascent RNAs and 7SK

snRNA^{184,185} functionally crosstalk with BRD4 and other pause release factors remains to be further investigated.

5. Post pause release: After pause release, Pol II resumes transcription typically in a highly processive manner until encountering roadblocks, such as positive supercoiling-generated DNA torsional stress, gene body-embedded nucleosomes, co-transcriptional mRNA processing events (e.g., splicing), and transcription-replication conflicts, that require additional factors to overcome these elongation barriers. DNA topoisomerase I, besides associating with promoter-engaged Pol II to facilitate PIC assembly and pause release, is also found in gene bodies extending to the transcription termination sites to help relieve DNA torsional stress in an enzymatic activity- and BRD4-dependent manner.¹⁸² Many of the BRD4-interacting proteins, e.g., 53BP1, BARD1, GLTSCR1, ATAD5, CHD4, JMJD6, and NSD3 (Figure 4B), possess chromatin-remodeling, histone-modifying, topologically-associating/scaffolding, or genome-surveillance activity to help Pol II pass through downstream roadblocks. CHD4 (chromodomain helicase DNA-binding protein 4), as a subunit in the NuRD chromatin-remodeling complex, interacts with the ET domain of BRD4 to promote fusion oncoprotein access to super-enhancers in rhabdomyosarcoma¹⁸⁶ where BRD4-L has recently been shown to act as an oncogenic factor⁹³ in contrast to its tumor-suppressive role in breast cancer.⁵⁴ Other examples include NSD3 (nuclear receptor-binding SET domain protein 3)-facilitated BRD4 association with CHD8 chromatin remodeler¹⁸⁷ and c-Myc oncoprotein,¹⁸⁸ as well as BRD4-ATAD5 (ATPase family AAA domain-containing 5)-regulated replication fork progression.^{29,30,148}

A specific class of BRD4-interacting proteins is involved in co-transcriptional RNA processing linked to Pol II elongation. This effector function of BRD4 mediated by direct protein-protein interaction (PPI) is distinct from the indirect transcriptional control of genes encoding RNA processing factors including select serine/arginine-rich splicing factors (SRSFs) and spliceosomal components.¹⁸⁹ Besides interacting with the RPB1 CTD,¹¹⁶ the cohesin (SMC1/SMC3/RAD21) complex,¹⁹⁰ and JMJD6¹⁶ that forms a unique splicing hub controlling c-Myc-promoted cancer metabolic programs,¹⁹¹ BRD4 also modulates mRNA processing via direct contact with specific components of the mRNA splicing machinery such as heterogeneous nuclear ribonucleoprotein M (hnRNPM) and FUS (fused in sarcoma; also known as hnRNPP2).²² From BRD4 isoform IP-proteomic analysis³³ performed in immortalized W12E keratinocytes, which are derived from a low-grade cervical lesion naturally harboring extrachromosomal HPV16 (ecHPV16) genomes, BRD4-L was indeed found to associate with JMJD6, RAD21, SRSF protein kinase 2 (SRPK2) and its substrate SRSF2,²⁴ as well as many mRNA splicing components involved in spliceosome assembly and function (Figure 4C, *green*), splicing regulation (*red*), and RNA binding via RNA recognition motif (RRM)-containing proteins implicated in m⁶A recognition (*blue*).³³

In contrast, BRD4-S, while not showing direct association with these RNA processing factors in W12E nuclear extracts, was specifically enriched in gene bodies in genome-wide binding profiling performed in MDA-MB-231 TNBC cells,⁵⁴ indicating that BRD4-S and BRD4-L may both be implicated in RNA processing via a mechanistically distinct and/or cell state-specific manner. This notion is supported by observations that (1) BRD4 inhibitor treatment consistently alters mRNA splicing patterns^{189,192} and RNA foci formation,¹⁹³ and

(2) BRD4-L-interacting RRM-containing YTHDC1 is involved in BRD4-regulated enhancer function,¹⁴¹ alternative spliceosome complex assembly, and splice site usage¹⁹⁴ depending on its reading of m⁶A-modified enhancer RNA or precursor mRNA.

Through interaction with distinct protein partners and ncRNA/pre-mRNA association, BRD4 controls multiple steps of chromatin dynamics, transcription, and co-transcriptional events. It is thus important to define the molecular mechanisms underlying BRD4-modulated proximity (i.e., short-range) and long/ultralong-range chromosomal organization and oncogenic ecDNA amplification, particularly the role of BRD4 protein isoforms and post-translational modifications in these processes.

Distinct BRD4 Isoform Function in Cancer Development and Viral Latency/Reactivation

Mechanistic understanding of BRD4 isoform function is crucial for identifying key gene targets and regulatory pathways/factors for the development of therapeutic options. During disease progression, BRD4, with its growth-promoting/immunomodulatory function, often drives proto-oncogene and immune checkpoint gene expression, as seen with enhanced levels of c-Myc in leukemia and high-grade serous ovarian carcinoma (HGSOC),^{79,80,82} FosL1/AP-1 in lung adenocarcinoma,¹⁹⁵ immunosuppressive programmed cell death protein 1 (PD-1) in T cells,¹⁹⁶ and programmed cell death ligand 1 (PD-L1) in cancer cells.¹⁹⁷ In most of these studies, the cancer-driving role of BRD4 is attributed to the more abundant BRD4-L, even though most of the experimental strategies involving pan-BRD4 RNAi or antibodies targeting the N-terminal region of BRD4 cannot readily distinguish the specific BRD4 isoform involved. This is particularly concerning, as a predominant BRD4-L degradation product comigrating with BRD4-S is often detected in SDS-PAGE and WB analysis. Without using isoform-specific antibodies (Figure 2) or conducting functional rescue with ectopically expressed isoforms, functional designation of BRD4-L and BRD4-S based purely on gel size detection could be misleading.

Recently, opposing functions of BRD4-L and BRD4-S in cancer development and viral latency/reactivation have been defined mainly by parallel studies of individual BRD4 isoform knockdown and subsequent rescue in the same experimental setting. The demonstration of BRD4-L as a tumor suppressor versus the oncogenic function of BRD4-S in breast cancer, particularly the TNBC subtype, is thus convincingly established via cell-based 2D/3D proliferation, migration, and invasion assays, as well as *in vivo* xenograft and transgenic mouse models⁵⁴ (Figure 5A). It appears that BRD4-S and BRD4-L both associate with acetylated nucleosomes and interact with the same TF, engrailed homeobox 1 (EN1), typically expressed in the brain but ectopically reactivated in TNBC cells. However, BRD4-S-upregulated matrisome (i.e., extracellular matrix-ECM network) gene expression through enhancer binding is distinct from the promoter association of BRD4-L, resulting in altered expression of matrisome genes encoding mucins (5AC, 5B), syndecans (SDC3, SDC4), type II cystatins (CST1, CST2, CST4), cathepsins (CTSB, CTSW, CTSZ), semaphorins (SEMA3B, SEMA3F, SEMA4D, SEMA6B), plexins (PLXNA2, PLXNA3, PLXND1), and many others (Figure 5A). This highlights the importance of *location*-dependent regulation of BRD4 isoform functions.

A pro-leukemia function of BRD4-S versus a potential anti-leukemia role of BRD4-L¹⁹⁸ in driving BCL2 and c-Myc oncogene expression has likewise been described in the mixed-lineage leukemia (MLL) fusion protein-positive (i.e., MLL-X, derived from chromosomal translocation/rearrangement) subtypes of acute myeloid leukemia (AML; Figure 5A). Similar BRD4-S-promoted colorectal cancer (CRC) cell proliferation/migration and tumor growth have also been described.⁹⁴ In both CRC and MLL-X AML, SRSF protein kinase 1 (SRPK1)-mediated phosphorylation of SRSFs that regulate splice site recognition and spliceosome assembly seems to enhance BRD4-S 3'-terminal exon 12 usage (Figure 3), thus favoring pre-mRNA splicing towards BRD4-S over BRD4-L. This is consistent with the observation that inhibition of SRPK1 kinase activity by small-molecule inhibitor SPHINX31 leads to increased BRD4-L (i.e., reduced BRD4-S) transcript and protein levels without altering total BRD4 transcripts. However, the specific regulatory SRSF proteins involved, e.g., SRSF1/2 in MLL-X AML¹⁹⁸ vs. DEAD-box RNA helicase DDX27-recruited SRSF6 in CRC,⁹⁴ may vary by the cancer type and its tumor microenvironment. Although knockdown of BRD4-L shows only small, if any, effects in both studies, a precise role of BRD4-L remains to be clarified. Nonetheless, a tumor-suppressive role of BRD4-L in preventing aging-associated cancer development in Hutchinson-Gilford progeria syndrome (HGPS), which is a premature aging disorder, has also been suggested based on oncogenic transformation resistance in HGPS patient-derived cell studies.¹⁹⁹

Context-dependent regulation is not constrained to the spatial locations where BRD4 isoforms bind to their regulated genes but also depends on subcellular compartmentalization, cell/tissue/organ-specific status, and the microenvironment. Unlike the opposing functions observed in breast cancer and MLL-X-positive AML, BRD4-L is the *de facto* oncogenic driver in rhabdomyosarcoma,⁹³ where BRD4-S acts as a gatekeeper (i.e., suppressor) to restrict BRD4-L function in upregulating invasion/migration-associated integrin *ITGA4/5* expression, particularly in fusion protein (FP)-negative embryonal rhabdomyosarcoma (FP⁻ ERMS; Figure 5B).

While antagonizing activities seem to be a common feature in several of the examined cancer types, BRD4-L and BRD4-S can also act cooperatively or synergistically in subsequent disease progression or in other biological contexts. When cancer cells metastasize to distal organs, the opposing activities of BRD4-L and BRD4-S observed in primary breast tumors switch to functional dependency or cooperativity during lung metastasis, since knockdown of either isoform significantly diminishes TNBC metastasis to the lungs in an MDA-MB-231 lung metastasis (LM)-prone mouse xenograft model.⁵⁴ This functional dependency/cooperativity of both BRD4 isoforms is also found in *BRD4* locus-amplified, c-Myc-driven HGSOC in which Neuregulin 1 (NRG1) important for activating membrane tyrosine kinase receptor signaling is upregulated by BRD4-L and BRD4-S acting together with the tumor-promoting activity of the SWI/SNF complex, with each isoform additionally regulating unique gene signatures, e.g., BRD4-L-enhanced E2F1 versus BRD4-S-elevated VEGF pathways²⁰⁰ (Figure 5C). This indicates that BRD4-L and BRD4-S can both play an oncogenic role by targeting common and unique cellular pathways during disease progression.

The recruitment of the SWI/SNF chromatin remodeler possessing tumor-promoting function by BRD4-S/L in HGSOE is distinct from a typically tumor-suppressive role of SWI/SNF, which in the case of human immunodeficiency virus type 1 (HIV-1) is recruited mainly by BRD4-S, and not by BRD4-L, to the downstream BRD4-bound +1 nucleosome.⁶⁹ BRD4-S-recruited SWI/SNF silences HIV-1 promoter activity during HIV-1 latency, particularly in CD4⁺ T cells, and likely suppresses endogenous retroelements in the host genome as well (Figure 5D). Reactivation of the latent HIV provirus can be achieved by treating cells with activating cytokines (e.g., TNF- α) or latency-reversing agents (LRAs) in the “shock-and-kill” regimen to displace the repressor complexes and/or remodel the chromatin environment. This triggers HIV transcription that can be further enhanced by BRD4-L-mediated recruitment of TFs (e.g., NF κ B or AP-1) for transcription initiation and P-TEFb for Pol II pause release, or by the HIV-encoded transcriptional activator Tat in complex with P-TEFb during elongation (Figure 5D). Of note, P-TEFb recruitment by Tat or by BRD4-L is mutually exclusive.¹²² This intricate partner interaction switch from HIV latency to reactivation elegantly illustrates a highly coordinated molecular action mediated by distinct BRD4 isoforms in modulating viral gene expression.

In HPV, viral latency is partly achieved by HPV-encoded full-length E2,^{13,201} double-spliced E2 C-terminal domain (dsE2C),¹³ E8^ΔE2C (later simplified as E8^ΔE2),²⁰² and E1M^ΔE2C²⁰³ transcriptional repressors that limit viral gene expression typically in the basal and parabasal epithelial layers. Full-length E2, in complex with BRD4-L and through binding to its cognate *cis*-elements in the E6 early promoter-proximal region, inhibits E6 and E7 oncogene expression primarily by blocking TFIID and Pol II entry in an acetylated nucleosome-dependent manner.^{13,134,204} This transcription-repressive function of E2 is separate from its transactivating role in gene regulation and its involvement, together with HPV-encoded E1 helicase, in initiating viral DNA replication,²⁰⁵ with functional distinction depending on specific DNA contexts and the partner proteins involved.^{206,207} When the repressor function of E2 is disrupted, often through viral genome integration into host chromosomes that partially truncates E1/E2-coding sequences, BRD4-L/S then works with cellular factors (e.g., AP-1)²⁰⁸ acting through the enhancer-containing HPV upstream regulatory region (URR) or integration-adjacent cellular enhancers to activate E6/E7 expression, leading to cellular hyperproliferation and sometimes oncogenic progression.^{13,206,207} Without integration, latent HPV genomes remain as episomal ecDNA and use NPS-phosphorylated BRD4-L/S to potentiate E1/E2-dependent early HPV replication.^{28,33} Upon epithelial cell differentiation, which is critical for viral capsid production and virion assembly, BRD4-L/S then recruits DDR factors, such as p53-binding protein 1 (53BP1), implicated in non-homologous end joining (NHEJ), and BRCA1-associated RING domain 1 (BARD1), involved in homologous repair (HR), to help overcome replication stress associated with high-copy episomal eHPV replication needed for the productive HPV infection cycle.³³ This often occurs in a temporal and spatial-dependent manner, first through BRD4-L-enhanced E2 binding to the HPV origin of replication (*ori*), followed by BRD4-L-mediated recruitment of 53BP1 and BRD4-S, with the latter further bringing in BARD1 during the early stage of differentiation-dependent HPV genome amplification (Figure 5E). A switch of DNA repair pathways from error-prone NHEJ mediated by BRD4-L/53BP1 to error-free HR facilitated by BRD4-L/BARD1 occurs at a subsequent replication stage to ensure faithful

genome replication,³³ which is likely triggered by reduced levels of histone H4 lysine 20 dimethylation (H4K20me2) needed for stable 53BP1 binding,²⁰⁹ thus dislodging 53BP1 from replicating HPV chromatin.

The recruitment of DDR factors through distinct BRD4 isoforms is crucial not only for alleviating the replication stress associated with productive viral genome replication but also for maintaining host genome integrity, whose disruption or instability/imbalance is often linked to carcinogenesis. DDR factors uniquely associated with BRD4-L, as identified by proteomic network analysis³³, include 53BP1, RIF1 and NSD2 (also known as WHSC1) in NHEJ; BARD1 and BRCA1 in HR; INO80, MCRS1 and ACTR8 in global genome nucleotide excision repair (GG-NER); TFIIH, CDK9, XAB2 and Pol II in transcription-coupled nucleotide excision repair (TC-NER); and RFC1/2/3, PCNA, CUL4A and XPC in both GG-NER and TC-NER pathways (Figure 5F).^{6,25,33} In contrast, BRD4-S and BRD4-S(b) show selective association with HR factors BARD1/BRCA1³³ and DDR-shielding condensin II components SMC2/NCAPD3³⁴, respectively. Besides direct effector function, BRD4 isoforms also indirectly modulate the levels of DDR factors through transcriptional control, as evidenced by BRD4-L-regulated gene transcription of *XRCC5/6*³⁶ and *CtIP*³⁷ involved individually in the NHEJ and HR repair pathways. Clearly, multi-tier regulation through protein effector function and transcriptional/posttranscriptional control is used by distinct BRD4 isoforms to modulate key steps in cancer progression and viral latency/reactivation.

BRD4 Phosphorylation and Other Post-Translational Modification (PTM)

The biological function of BRD4 isoforms lies not only in effector association and transcriptional regulation but is further dictated by post-translational modifications. PTMs can generate diversity in BRD4 protein isoforms by creating new PPI interfaces, altering protein stability, modulating enzymatic activities, and/or changing subcellular localization crucial for cancer/disease progression and viral latency/reactivation, making PTM sites potential biomarkers in targeted therapies. Reported PTMs in BRD4 include phosphorylation, methylation, acetylation, ubiquitination, sumoylation, and hydroxylation, which are typically identified by mobility shifts in SDS-PAGE, bioinformatic and computational analyses, *in vitro* modification assays, or highly sensitive and quantitative liquid chromatography electrospray ionization tandem mass spectrometry (LC-ESI-MS/MS) or liquid chromatography-matrix-assisted laser desorption/ionization (LC-MALDI) MS analysis. The identified PTMs are typically validated by site-directed mutagenesis and/or PTM-specific antibody-based detection (e.g., WB and cell imaging). Among these, the functional effects of BRD4 phosphorylation have been most extensively studied.

Phosphorylation of BRD4 was first observed by a mobility difference in SDS-PAGE between bacteria- versus insect cell-expressed recombinant protein fragments containing either NPS or CPS. Analysis of amino acid sequences identified seven putative casein kinase II (CK2) consensus sites (pS/TxxE/D; x is any amino acid) in the NPS that were subsequently confirmed by dephosphorylating pNPS in insect cell-purified protein fragments with lambda or calf intestinal alkaline phosphatase and by phosphorylating NPS in bacteria-purified recombinant fragments with CK2¹⁷ (Figure 6A). An additional

phosphorylation site (T500) was independently identified by MS analysis. CK2 is a cell-intrinsic NPS kinase in both normal and cancer cells, as perturbation of CK2 expression and its kinase activity in keratinocyte-derived Ker-CT cells, primary neurons, and SUM159 TNBC cells by CK2 knockdown or inhibitor treatment shows reduced NPS phosphorylation of endogenous BRD4 proteins.^{17,83,125} Phosphorylation of NPS, particularly at S492 and S494, is critical for unmasking the chromatin-binding activity of BD1/2 and for BRD4-mediated transcriptional regulation typically via phosphorylation-dependent association with sequence-specific transcription factors and other pNPS interactors (see Figure 4B). This NPS region is conserved in all BET family proteins.¹⁷ Thus, NPS phosphorylation likely plays an important role in BRD2, BRD3, and BRDT function as well. Besides CK2, other kinases, such as protein kinase A (PKA),¹⁷ I κ B kinase ϵ (IKK ϵ),¹⁷ and CK1 δ ,²¹⁰ can also phosphorylate the NPS, with CK1 δ -phosphorylated BRD4 shown to be critical for cerebellar granule cell progenitor function and neurogenesis.²¹⁰

Phosphorylation in other regions of BRD4 has also been described (Figure 6A). Hyperphosphorylation of BRD4-L at T249/S1045/S1117/T1186 by cyclin-dependent kinase 1 (CDK1)/cyclin B occurs during mitosis.²¹¹ The CDK1-phosphorylated T1186 is recognized by polo-like kinase 1 (PLK1), which then phosphorylates BRD4-L at S24 and S1100 (Figure 6A), leading to recruitment of the APC/C^{CDH1} E3 ubiquitin ligase and proteasome-mediated degradation of BRD4-L.⁵⁶ This mitosis-specific phosphorylation cascade (CDK1-PLK1-APC/C^{CDH1}) could potentially reduce the elevated BRD4 protein levels observed in BBD β -resistant prostate cancer cells, thus resensitizing cancer cells to therapeutic drug treatment. Interestingly, T1186 and additionally T1212 are also phosphorylated by c-Jun N-terminal kinase (JNK), which triggers BRD4-L release from chromatin²¹² and likely results in BRD4-L dissociation from metaphase chromosomes.⁵⁰ This could partly explain why BRD4-L may not be a mitotic bookmarking factor.⁵³ JNK-mediated phosphorylation seems to inactivate the reported HAT activity of BRD4 and in turn switch on its kinase activity, converting BRD4-L from a chromatin binder to a promoter regulator, which likely helps Pol II release from promoter-proximal pause sites through enhanced phosphorylation of Pol II CTD at Ser2, CDK9 at Thr186, and c-Myc at Thr58.²¹² It would be of great interest to examine whether BRD4-S, lacking S1100/T1186 found in BRD4-L, is in fact the authentic mitotic chromosome-bookmarking factor that provides a binding platform for viral genome tethering and conveys transcriptional memory upon cell reentry to the G1 phase.

A key event in phosphorylation regulation is recognition of the modified phosphoresidues by reader proteins that trigger specific downstream signaling events. A BRD4-interacting protein, peptidyl-prolyl isomerase NIMA-interacting 1 (PIN1), is a peptidyl-prolyl *cis-trans* isomerase (PPIase) shown to bind pS/T-Pro (P) motifs, such as pT204-P205 of BRD4,²¹³ and functions as a post-phosphorylation regulatory factor to regulate substrate conformation. Although the pT204 kinase has not yet been identified, PIN1 binding to pT204, which is unique to BRD4 and not found in BRD2 and BRD3, converts pT204-P205 from the *cis* to *trans* configuration, thus enhancing BRD4 protein stability by reducing polyubiquitination and further stimulating BRD4 interaction and recruitment of CDK9, target gene transcription, and tumor-promoting function in gastric cancer cells.²¹³

Other phosphorylation sites that also confer BRD4 protein stability are tyrosine (Y) 97 and Y98 in BD1 of BRD4, as phosphorylation of both Y97 and Y98 by Janus kinase 2 (JAK2) in response to cytokine IL6/IL8 secreted by cancer-associated fibroblasts (CAFs) in the tumor microenvironment of CRC enhances pBRD4 interaction with ubiquitin C-terminal hydrolase L3 (UCHL3), a BRD4 deubiquitinase highly active in CRC.²¹⁴ UCHL3 is distinct from the earlier identified deubiquitinase DUB3¹³² that counteracts the poly-ubiquitinating activity of a BRD4 E3 ubiquitin ligase containing Cullin 3, RING, and BRD4-binding adaptor SPOP (speckle-type POZ protein).^{215–217} While Y97 crucial for BBD β binding is conserved among BET proteins, Y98 is only present in BRD4, but not BRD2 or BRD3, making BRD4 a unique substrate regulated by extrinsic IL6/IL8-JAK2 signaling in promoting cancer cell expansion. Importantly, phosphorylation at Y97/Y98 not only confers drug resistance in cancer cells but also enhances BRD4 binding to chromatin and interaction with JAK2-activated transcription factor STAT3 (signal transducer and activator of transcription 3) at growth-promoting enhancer/super-enhancer regions.²¹⁴ This JAK2-dependent phosphorylation at Y97/Y98 is distinct from CK2-enhanced NPS phosphorylation in stimulating BRD4 chromatin association. It would be of great interest to examine whether these two phosphorylation events are interlinked or functionally independent.

Clearly, distinct kinases (CK2, PKA, IKK ϵ , CK1 δ , CDK1, PLK1, JNK, and JAK2), acting in a context-dependent manner, create a combination of phosphorylation codes that trigger specific downstream signaling pathways to modulate cellular activities that are either uniquely (e.g., C-terminal modifications) or commonly (e.g., NPS phosphorylation) modulated by distinct BRD4 isoforms (Figure 6B). The degradation and chromatin release of BRD4-L, triggered by distinct phosphorylation events at its unique C-terminal extension, likely result in a dynamic BRD4-L turnover through cell cycle progression. In contrast, BRD4-S, albeit less abundant than BRD4-L in most cells, provides a more stable form in response to intra- and extracellular signaling events. The functions of the mostly cell type-specific and stress-induced BRD4-S(b) are yet to be defined.

Besides phosphorylation, BRD4 is also modified by ubiquitination (Ub), sumoylation (SM), acetylation (Ac), methylation (Me), and hydroxylation (HO), based mostly on software prediction and/or mass spectrometry analysis, and some of the PTMs have been experimentally validated (Figure 6C). Ubiquitination of BRD4 by the SPOP E3 ubiquitin ligase, first described in prostate cancer,^{215–217} and deubiquitination by DUB3¹³² in prostate cancer and UCHL3²¹⁴ in colorectal cancer mainly regulate BRD4 protein stability. Conjugation of a small ubiquitin-like modifier (SUMO) protein to the ϵ -amino group of a lysine residue, analogous to ubiquitination, is another way to control protein stability and transcription activity, among other biological functions.^{218,219} Among multiple predicted sites, sumoylation at K1111 of BRD4 by E3 SUMO ligase chromobox homolog 4 (CBX4) is reported to stabilize BRD4 by antagonizing polyubiquitination, thus enhancing the expression of proinflammatory cytokines such as IL-1 β , TNF- α , and IL-6 in synovial fibroblasts.²²⁰

Acetylation of BRD4 at K332 by PCAF (p300/CBP-associated factor; i.e., KAT2B) has been shown to enhance BRD4 association with TF intestine-specific homeobox (ISX) that promotes epithelial-mesenchymal transition (EMT) and lung cancer initiation/progression

through the PCAF-ISX-BRD4 axis-enhanced expression of EMT-promoting TFs, such as TWIST1 and SNAIL1.¹⁰⁴ Interestingly, ISX in complex with BRD4 also depends on PCAF-acetylated K69 in ISX for binding specifically to BRD4 BD2 (not BD1), similar to that described for TIP60-acetylated K73/K76 in TWIST1, which likewise contact uniquely BD2 to control BRD4-regulated EMT in basal-like breast cancer cells,¹⁰³ and CBP-acetylated K146/K187 in SNAIL1 that are critical for BRD4-promoted gastric cancer cell invasion and metastasis.²²¹

In addition to acetylation, BRD4 can also be methylated at lysine residues by certain SET domain-containing lysine methyltransferases (KMTs) or at arginine residues by specific protein arginine methyltransferases (PRMTs). Mono-methylation of BRD4 by SET domain-containing protein 6 (SETD6) at K99 within BD1, which is also conserved in BRD2 and BRD3 but not BRDT, does not affect BD1 binding to acetylated chromatin but reduces BRD4 interaction with TF E2F1, leading to selectively reduced ribosomal gene transcription thus underlying SETD6-mediated BRD4 methylation in inhibiting protein translation.²²² Unlike the inhibitory K99 methylation, PRMT1-mediated asymmetric dimethylarginine (ADMA) formation at R179, R181, and R183 residues of BRD4, which are situated downstream of BD1 and not conserved in other BET proteins, is critical for BRD4-promoted ovarian cancer cell migration, invasion, and tumorigenesis.²²³ Methylation of these three arginine residues, counteracted by the JMJD6 demethylase and recognized by the Tudor domain-containing SPF30 reader protein, increases NPS phosphorylation needed for BRD4 binding to targeted chromatin and is functionally linked to TGF- β signaling.²²³ This PTM crosstalk between arginine methylation and NPS phosphorylation is also important for partial EMT and TGF- β 1-induced kidney fibrosis, likely through PRMT1-enhanced BRD4 methylation and NPS phosphorylation that, besides regulating BRD4 target gene transcription, additionally modulates SNAIL1 acetylation and protein stability through direct protein-protein interaction.²²⁴ Surprisingly, DNA damage-induced arginine mono-methylation, and not the typical asymmetric di-methylation, at the same R179/R181/R183 residues by the PRMT2/PRMT4 complex likewise enhances BRD4 binding to acetylated chromatin and upregulates the expression of cell cycle, chromatin remodeling, and DNA repair genes in breast cancer cells, thus facilitating BRD4-mediated DNA repair and mouse xenograft mammary tumor growth.²²⁵

Besides these canonical PTMs, BRD4 is found to be hydroxylated by the 2-oxoglutarate dioxygenase activity of JMJD6 at 19 lysine residues,²²⁶ of which 14 are situated in the positively charged BID (Figure 6C) that is important for transcription condensate formation,¹¹³ nucleosomal DNA binding,¹⁴⁵ and intramolecular contact with the negatively charged pNPS to control factor recruitment and the acetyl-chromatin binding activity of BRD4.¹⁷ This oxygen level-dependent lysine hydroxylation, mostly detected in the BID, is also found in BRD2 and BRD3, although the extent of modification in BRD2 (four lysines) and BRD3 (six lysines) is significantly less than that found in BRD4. It would be of great interest to define whether any of these lysine-hydroxylated residues uniquely identified in BRD4 would alter the functional properties of BRD4 versus BRD2 and BRD3, e.g., the cellular distribution of BRD4 hydroxylated at K286/K289/K291 residues situated in the nuclear localization signal¹¹⁴ and whether K332-hydroxylated BRD4 blocks PCAF-

mediated BRD4 K332 acetylation, acetyl-K332-dependent BRD4 interaction with ISX, and PCAF-ISX-BRD4 axis-regulated EMT and lung cancer progression.¹⁰⁴

Targeting IDR for BRD4 Inhibition

The extensive PTMs identified in BRD4 particularly at the negatively charged NPS (phosphorylation) and positively charged BID (hydroxylation), both situated within the internal IDR, suggest that NPS and BID, via intra/inter-molecular contact switches and combinatorial PTM fine-tuning, play a key role in regulating BRD4 dynamics and functional properties.¹³⁶ This internal IDR with three charged clusters (NPS, BID, CPS) is distinct from the C-terminal proline-rich IDR (see Figure 2). A general feature of IDRs, often referred to as low complexity domains (LCDs) or low complexity regions (LCRs), is the conformational adaptability to form weak multivalent interactions with other IDRs or chemically compatible surfaces, whether unstructured or structured. The interface can be formed through hydrogen bonding, charge-charge, hydrophobic or π - π interactions, among others, to mediate assembly of diverse protein networks/hubs, foci/puncti, biomolecular condensates, liquid-liquid phase separation (LLPS), or liquid/hydrogel-like membrane-less compartments.^{227–233}

In drug development, folded domains (e.g., kinase, HDAC, bromodomain, etc.) are ideal targets for obtaining high-affinity compounds at nano/picomolar ranges through primarily structure-guided chemical modification. These pocket inhibitors, however, frequently exhibit off-target inhibition of other structurally and/or functionally related protein domains, incurring strong side effects and general toxicities (Figure 7A, *left*). On the other hand, targeting IDRs provides an innovative approach to modulate a contact interface created through cooperative multivalent, low-affinity interactions, often leading to a transition from a disordered-to-ordered structure that is distinct, highly selective, and not typically found in other functionally related proteins, thus significantly minimizing off-target inhibition and potential side effects (Figure 7A, *right*).

With the unsatisfactory results of BBD*i* in clinical trials,^{4,9} identification of non-bromodomain inhibitors offers new hope for BRD4-directed therapies. Indeed, a new class of PPI inhibitors (PPI*i*) targeting pNPS in the internal BRD4 IDR that modulates chromatin accessibility of BD2/BD1, factor recruitment, and intra/inter-domain interaction has been developed.^{28,33} The identification of IDR-targeting PPI*i*, termed IDR*i*, is based on the phospho-switch mechanism in which acetylated chromatin-binding activity of BD1/BD2 is regulated by the phosphorylation state of NPS.¹⁷ When non-phosphorylated, the NPS masks BD2 and prevents BD1/BD2 from binding to acetylated chromatin via NPS-BD2 and BD2-BD1 interactions. Phosphorylation of NPS by CK2 switches intramolecular NPS contacts from BD2 to downstream BID, leading to activation of BD2/BD1 chromatin-binding activity and concurrently recruiting an active form of p53 to its target genes (Figure 7B). It is important to note that protein-protein contacts may not necessarily lead to active transcription, as only pNPS-recruited p53 is active in target gene regulation whereas BID-associated p53 is inactive in DNA binding.¹⁷ It should also be mentioned that besides the intramolecular contact switch, NPS phosphorylation could also modulate inter-molecular BRD4 dimerization, partly determined by the ionic strength

of the solution (e.g., 250 mM NaCl).¹¹¹ The phosphatase involved in this phospho-switch has been identified as protein phosphatase 2A (PP2A), whose downregulation leads to increased BRD4 phosphorylation that correlates with TNBC and lung adenocarcinoma progression and drug resistance^{83,90,234} as well as neurodevelopment/neurodegeneration such as cerebellar neurogenesis,²¹⁰ mouse learning/memory,^{125,235} cocaine addiction,²³⁶ and memory impairment in fragile X syndrome.¹⁸ As NPS phosphorylation represents the active form of BRD4 for chromatin regulation and factor recruitment, it is conceivable that pNPS is broadly involved in BRD4 biology and thus provides a targeted region for anticancer, anti-inflammatory responses, and antiviral therapeutics development.

Through one-bead, one-compound microscopic screening of a 38,500 peptoid (i.e., amino acid-like N-substituted oligoglycines) library, two peptoids (DC-1: Nlys-Nlys-Nmea-Nlys-Ntyr-Nleu, and DC-2: Nlys-Nlys-Nffa-Nlys-Nleu-Napp) that bind specifically to pNPS, but not to non-phosphorylated NPS or another conserved CK2-phosphorylated region downstream of the ET domain (i.e., pCPS; see Figure 2), were identified (Figure 7C).²³⁷ DC-1, with half-maximal inhibitory concentration (IC_{50}) = 50 μ M, has a 2-fold higher affinity than DC-2 (IC_{50} = 125 μ M) for binding to the high-risk HPV16-encoded E2 (16E2) protein and effectively blocks pBRD4-dependent 16E2 binding to its cognate sequence situated ~1 kb upstream of the matrix metalloproteinase-9 (*MMP-9*) TSS in proliferating keratinocytes. It also diminishes pNPS-driven and 16E2-enhanced NF κ B binding to the *MMP-9* promoter-proximal (−0.6 kb) NF κ B binding site in differentiated keratinocytes, thus inhibiting BRD4-regulated *MMP-9* gene transcription.²⁸ Moreover, DC-1 and DC-2 both suppress pNPS-dependent transient HPV replication,³³ highlighting the proof of concept in developing IDR*i* with protease/peptidase-resistant and cell-permeable protein-binding ligands.

Although peptoids are highly selective ligands, their cell permeability and affinity toward target protein binding are generally low, which potentially thwarts further development of the initial hits. Through extension to large-scale high-throughput chemical library screening of 200,000 pharmacologically expandable small molecules, 83 compounds, of which 37 are commercially available, were identified to alleviate BRD4-E2 repressor complex-mediated inhibition of an HPV enhancer/promoter-driven luciferase reporter activity. One of the hits, compound **14**, and its chemical derivative **90** show potent inhibition of HPV genome replication not only in differentiated keratinocytes but also in a three-dimensional (3D) HPV epithelial cell differentiation (i.e., raft) system mimicking the natural HPV infection life cycle without altering tissue integrity and epidermal differentiation.³³ Both compounds exhibit significantly improved potency compared to peptoids, with low to sub-micromolar IC_{50} (**14**: 1.5–5.1 μ M; **90**: 0.4–1.6 μ M) for inhibition of pNPS/pBRD4 interactions with viral 16E2, p53 and DDR factors 53BP1 and BARD1 (Figure 7D). Importantly, while effectively blocking HPV DNA replication, **14/90** have minimal perturbation of the genome-wide transcriptome and the chromatin landscape in HPV-infected keratinocytes, suggesting that the high selectivity of IDR*i* for a specific PPI interface does not disrupt global gene transcription and chromatin organization at least in a virus-infected cellular environment.

Another IDR*i*, SDU-071, modified from compound **21** identified from the 200,000-compound library screen, has significantly improved IC_{50} (2.7–3.1 μ M), compared with the

parent compound (40.0–49.6 μM), for inhibition of pNPS-dependent interaction with full-length p53 or its C-terminal regulatory region (aa 309–393) and strongly suppresses MDA-MB-231 TNBC cell growth, invasion, migration, and orthotopic mouse xenograft mammary tumor growth.²³⁸ SDU-071 and compound **21** both preferentially inhibit BRD4 interaction with p53 over 16E2. These findings indicate that IDR*i* can block concurrent or sequential protein interactions with select phosphoresidues within a conformationally dynamic 20-amino acid region. Conceivably, pNPS can create a “combinatorial phosphorylation code” by using different positions/numbers of the eight phospho-S/T residues to contact different interactors such as TFs p53, NF κ B, and AP-1, replication factors HPV E2 and E1, DDR factors 53BP1, BARD1, and other yet-to-be-identified chromatin regulators (Figure 7E). This is analogous to the CTD phosphorylation code²³⁹ that creates diverse signals and binding platforms within the heptapeptide repeats of RPB1 for distinct factor association with either non-phosphorylated or phosphorylated Pol II during PIC assembly and the initiation, elongation, and termination of transcription, along with co-transcriptionally regulated 5' capping, mRNA splicing, and 3' mRNA processing events (see Figure 4A).

Besides the internal IDR, the C-terminal proline-rich IDR unique to BRD4-L may offer greater selectivity for BRD4-L-specific targeting. A natural product PCG (procyanidin C-13,3',3''-tri-O-gallate), acting as an HIV latency-reversing agent through promoting Tat-dependent HIV transcription²⁴⁰ (see Figure 5D), has been shown to bind directly to either or both proline-rich segments (P1: aa 751–800, P2: 951–1031) of BRD4-L (see Figure 2) and convert BRD4-L nuclear condensates into static aggregates, thus inhibiting c-Myc and select target gene transcription.¹¹⁹ Interestingly, some histidine residues embedded within the P1 and P2 segments also play a key role in conferring the intracellular pH-sensing activity of BRD4-L in regulating macrophage-mediated inflammatory responses.²⁴¹

Other examples of IDR targeting include (1) compound EPI-7386 (masofaniten) that binds androgen receptor N-terminal AF-1 IDR, with $\text{IC}_{50} = 0.421 \mu\text{M}$ in inhibiting the prostate-specific antigen (PSA) luciferase reporter activity in LNCaP prostate cancer cells, and exhibits greatly reduced toxicity in clinical trials against metastatic castration-resistant prostate cancer;^{242,243} (2) compounds 10074-G5 and 10058-F4 that bind distinct IDR regions of the monomeric c-Myc bHLHZip domain (aa 363–381 and 402–412, respectively), preventing c-Myc/Max heterodimerization with $\text{IC}_{50} = 0.8\text{--}7.0 \mu\text{M}$, and induce myeloid differentiation in AML cells as well as neuronal differentiation and lipid accumulation in MYCN-amplified neuroblastoma cells;^{244,245} (3) compounds Nutlin-3 and RG7112 that bind MDM2 and disrupt its interaction with the p53 N-terminal transactivation domain IDR with $\text{IC}_{50} = 0.018\text{--}0.3 \mu\text{M}$, effectively inhibiting cell growth and viability of cancer cells expressing wild-type but not mutant p53;^{246,247} (4) compounds lipoamide and lipoic acid that disrupt FUS/TDP-43 condensates with $\text{IC}_{50} = 13\text{--}17 \mu\text{M}$ and alleviate neuronal defects caused by Amyotrophic Lateral Sclerosis (ALS)-associated FUS and TDP-43 mutants;²⁴⁸ and (5) compound A-485 that, through binding to the HAT domain of p300/CBP and at 10 μM , reduces p300 IDR-mediated LLPS and transcriptional capacity.²⁴⁹ Clearly, IDR targeting is feasible and represents a new direction for effective drug development that may minimize the strong side effects commonly encountered with pocket domain targeting during clinical trials and patient treatments.

Conclusions

Transcription is a highly coordinated event that requires RNA polymerases working in conjunction with an array of general and gene-specific transcription factors and cofactors to regulate distinct steps of the transcriptional process. This complexity of gene regulation is further complicated by the chromatin environment that adds another layer of regulation for TF access to enhancer/promoter regions. BRD4, as an epigenetic and transcriptional regulator, is a key factor bridging chromatin and DNA-driven processes by either acting as a scaffold/adaptor protein for locus-specific enhancer-promoter communications and long/ultralong-range chromatin interactions or as a checkpoint modulator controlling transcription progression and factor exchange. Three features underlying the action of a universal chromatin regulator for functioning as a gene/locus-specific factor are (1) post-translational modification, (2) isoform switch, and (3) partner association.

The advances in multi-omics and computational tools allow network analysis and pathway identification through loss- and gain-of-function approaches, along with functional rescue, to identify specific factors and/or protein domains/regions crucial for their action and regulation. Besides genetic dissections, pharmacological agents with the use of target-selective chemical inhibitors and/or PROTAC (proteolysis targeting chimera) degraders further expand the repertoire of toolkits available for biological studies. In the case of BRD4, both genetic and chemical analyses have identified the unique and shared properties of universally expressed long (BRD4-L) and short (BRD4-S) isoforms, extending beyond the seemingly stress- and cell type-specific function of BRD4-S(b). While broad perturbation of BRD4 or BET protein functions using non-selective genetic or chemical inhibition makes it difficult to unambiguously pinpoint the specific factor involved, functional rescue by unique isoforms or other family members helps to identify the implicated entities and pathways.

Among BET family proteins, the C-terminal proline-rich (sometimes referred to as HPQ-rich for histidine/proline/glutamine-rich) region is unique to BRD4-L. Thus, antibodies and genetic/chemical perturbations targeting epitopes or structural features residing in this IDR most likely reflect the functional properties of BRD4-L. Any reagents disrupting the functions of evolutionarily conserved N-terminal regions would require additional biological validation with the specific rescue factor(s) to designate the activity to unique BET family members (BRD4-L, BRD4-S, BRD3, BRD2). Moreover, target identification from *in vitro* analysis may not be fully validated by *in vivo* characterization, as the presence or absence of other cellular or viral proteins or macromolecules may mask or unmask the epitopes or surface features.

While the internal IDR-targeting compounds (e.g., **14** and **90**) appear to bind pNPS of full-length proteins or an IDR-only segment derived from purified recombinant BRD4-L, BRD4-S, BRD3, and BRD2 with comparable affinities *in vitro*, these compounds do show *in vivo* selectivity of specific BET proteins or isoforms in a context- and/or cell type-specific manner, presumably due to the presence of interacting partners that modulate the extent of surface exposure through direct complex formation and/or allosteric regulation. Comparison between *in vitro* and *in vivo* results is thus critical to provide a comprehensive

and unbiased view of BET protein action and compound selectivity. It should also be mentioned that although off-target inhibition and general toxicity are often seen with BBD or other folded domain inhibitors and PROTAC degraders, the strong side effects can be potentially alleviated through development of domain (e.g., BD1 or BD2 only)-selective inhibitors, allosteric inhibitors, dual-target inhibitors, or combination therapy with much reduced concentrations. Gain- and loss-of-function mutations uniquely found in the folded domains can also be explored for developing mutant-selective inhibitors/modulators or antibodies that do not cross-react with other structurally or functionally related wild-type proteins.

Beyond conventional wisdom in therapeutics design and optimization, targeting IDRs that are uniquely adapted to the partner-interacting interface is an innovative approach to circumvent the strong side effects encountered with domain-targeting inhibitors. In cases where BET members work jointly to control a biological process – e.g., in differentiation-dependent HPV genome amplification where BRD4-L, BRD4-S and BRD3 are all functionally involved,³³ in BRD4-amplified HGSOC where both isoforms contribute to oncogenic c-Myc and NRG1 pathways (Figure 5C), and in breast cancer metastasis in which BRD4-L/S are both critical for lung metastasis⁵⁴ – pNPS IDR-targeting and pan-BET inhibitors may be particularly effective in treating these diseases. However, in a context where BET members or BRD4 isoforms exhibit opposing activities such as in the TNBC and RMS tumor development, this non-selective pan inhibition may blunt the therapeutic efficacy by simultaneously blocking both oncogenic and protective functions.

The success with phosphoresidue-based IDR*i* can be likewise applied to other PTM residues in therapeutics development. Moreover, enhancement of drug modeling algorithms with the aid of artificial intelligence (AI) to include IDR/PTM information based on biochemical/biophysical and protein structure studies will be beneficial. With more drug-like IDR modulators and in-depth molecular understanding of protein function and compound selectivity facilitated by integrated genomic technology and computational AI approaches, more effective treatment regimens tailored to individuals based on their unique genetic and molecular profiles (i.e., personalized medicine) can be achieved for patients with viral, cancer, or inflammation-associated diseases in BRD4-targeted therapy.

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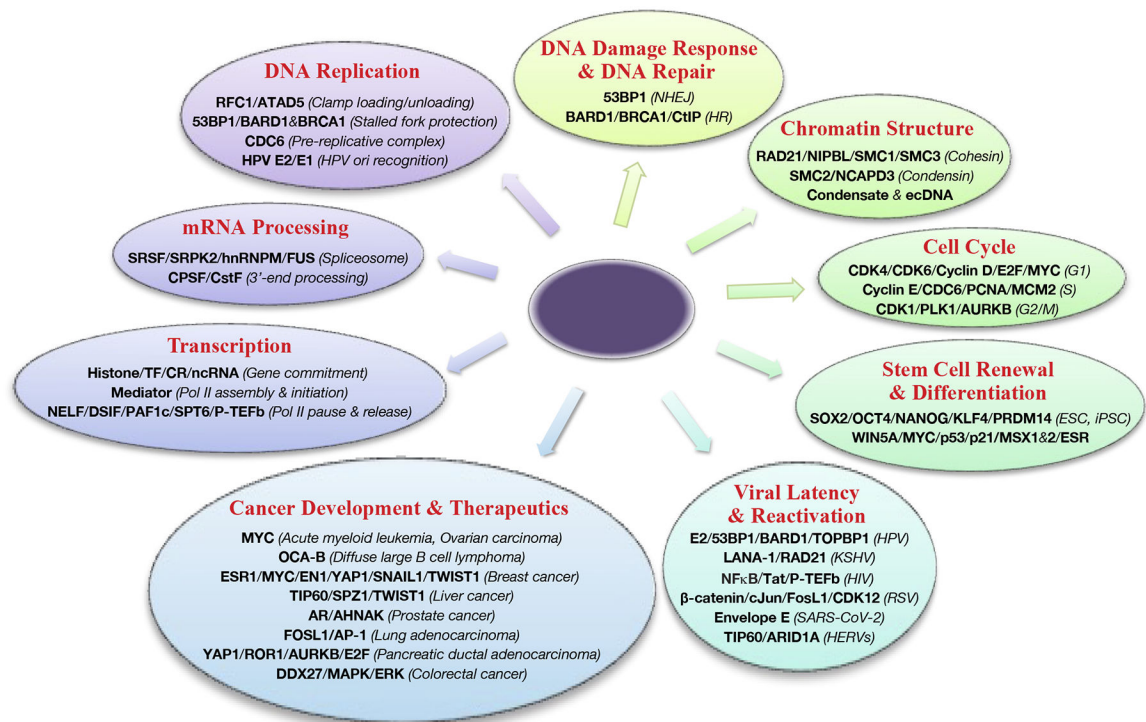


Figure 1. BRD4-Regulated Cellular Processes

BRD4 regulates transcription, mRNA processing, DNA replication, DNA damage response and repair, chromatin structure, cell cycle, stem cell renewal and differentiation, and viral latency and reactivation, as well as cancer development and therapeutics. Representative BRD4 gene targets, pathways, and interactors are indicated for each biological process. Abbreviations: TF, transcription factor; CR, chromatin regulator; ori, origin of replication; NHEJ, non-homologous end joining; HR, homologous recombination; ESC, embryonic stem cell; iPSC, induced pluripotent stem cell; HPV, human papillomavirus; KSHV, Kaposi's sarcoma-associated herpesvirus; HIV, human immunodeficiency virus; RSV, respiratory syncytial virus; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; HERV, human endogenous retroviruses.

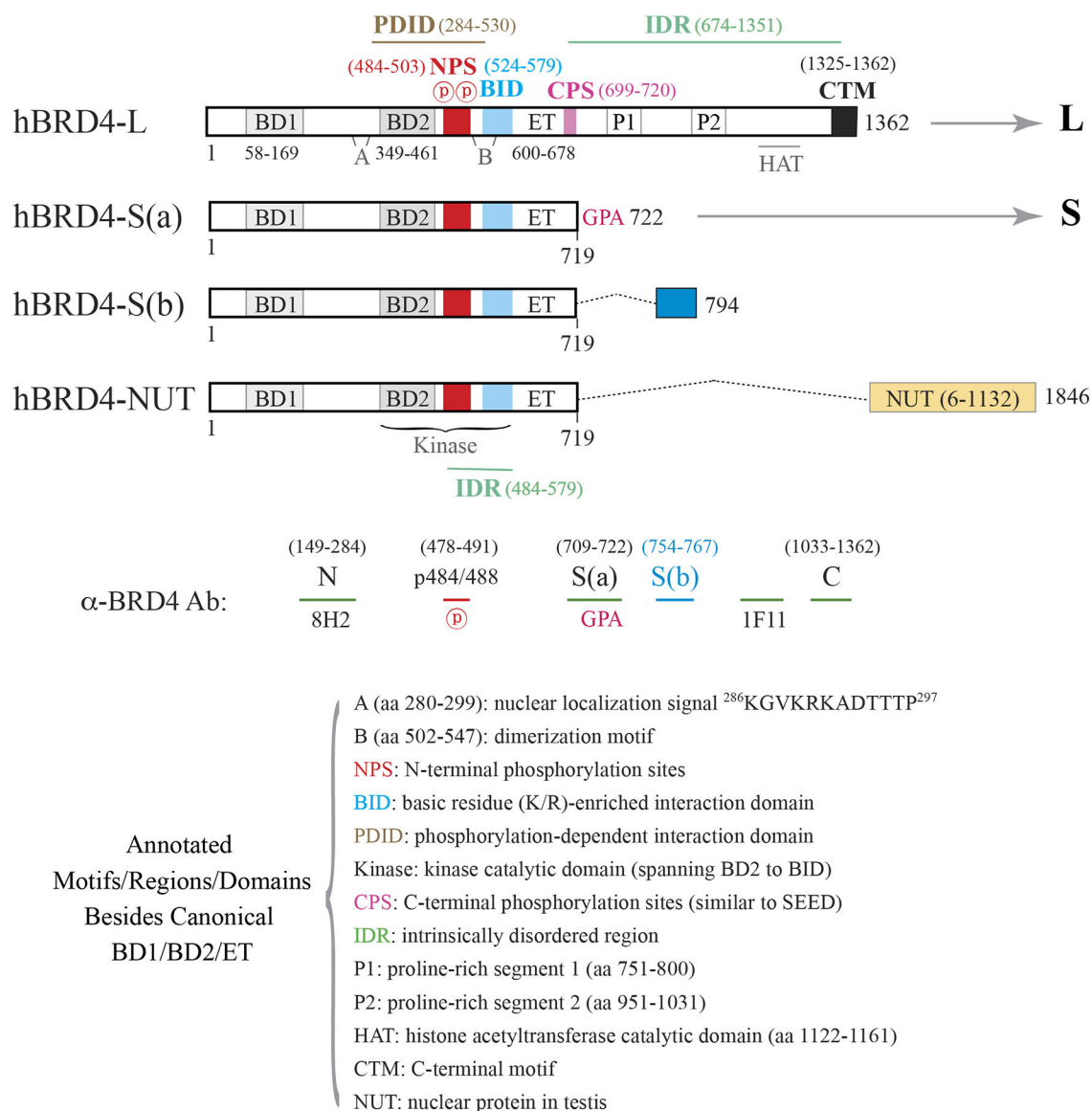


Figure 2. Domain Features of BRD4 Protein Isoforms

Schematic illustration of three human BRD4 protein isoforms: BRD4-L, BRD4-S(a), and BRD4-S(b), representing long (L) and short (S) isoforms a and b encoded by transcripts differing in 3' alternative splicing, as well as BRD4-NUT fusion protein, derived from t(15;19) chromosomal translocation. BRD4-S(a) is a ubiquitously expressed short isoform frequently referred to as BRD4-S. Anti-BRD4 antibodies recognizing specific isoforms and phosphorylated serine 484 and/or serine 488 (p484/488) are indicated, along with pan N and 8H2 antibodies recognizing a common N-terminal region. All antibodies are rabbit polyclonals, except 8H2 and 1F11 which are mouse monoclonal antibodies. Numbers mark the positions of amino acid (aa) residues. Annotated domains/regions include BD1 (1st bromodomain), BD2 (2nd bromodomain), NPS (N-terminal phosphorylation sites), PDID (phosphorylation-dependent interaction domain), BID (basic residue-enriched interaction domain), kinase catalytic domain (spanning BD2 to BID), Motif A (aa 280–299

containing the nuclear localization sequence ²⁸⁶KGVKRKADTTTP²⁹⁷), Motif B (aa 502–547, dimerization motif), ET (extra-terminal domain), CPS (C-terminal phosphorylation sites, coinciding with SEED), IDR (intrinsically disordered region), P1 (proline-rich segment 1 spanning aa 75–800), P2 (proline-rich segment 2 spanning aa 951–1031), HAT (histone acetyltransferase catalytic domain spanning aa 1122–1161), CTM (C-terminal motif), and NUT (nuclear protein in testis).

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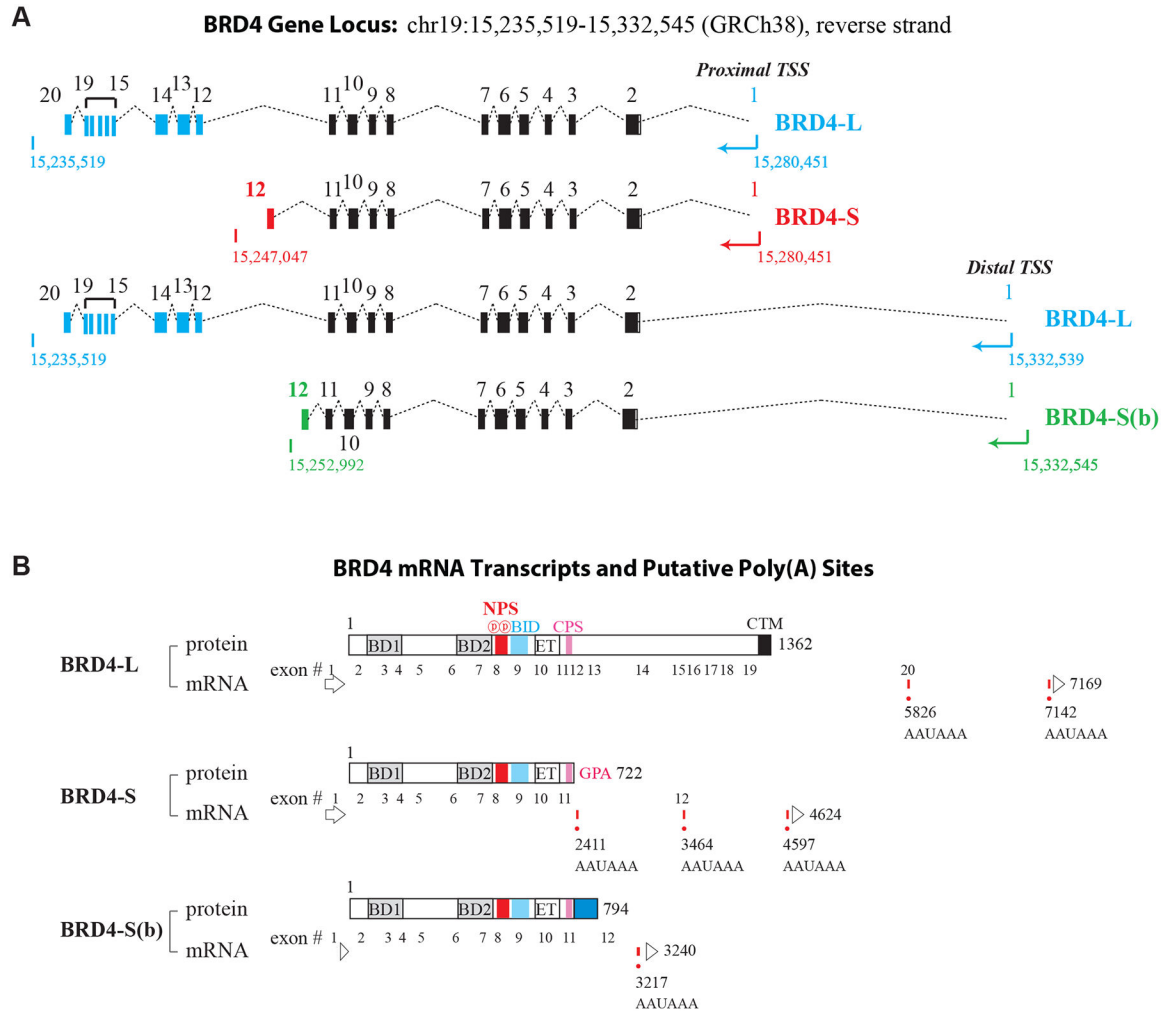


Figure 3. Human BRD4 Gene Organization and Alternative mRNA Transcripts

(A) BRD4 gene locus. Numbers are nucleotide positions designated by Ensembl Genome Browser based on human GRCh38. Arrow, transcription start site with transcribed direction. Filled box, coding exon. Empty box, 5' or 3' untranslated region (UTR). Dashed line, intron. (B) BRD4 mRNA and potential poly(A) sites. Grey arrow, coding exons. White arrow, 5' or 3' UTR. Red line and dot, position of AAUAA polyadenylation signal.

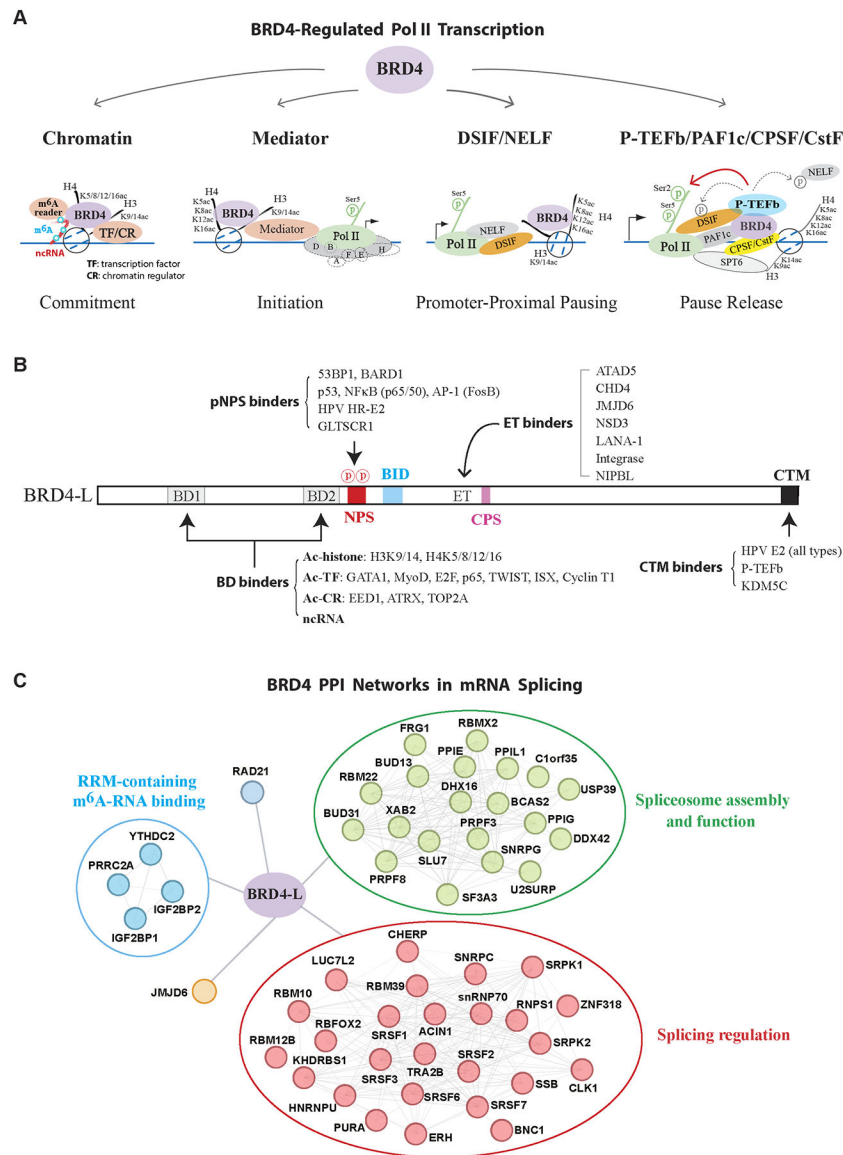


Figure 4. BRD4-Regulated RNA polymerase II (Pol II) Transcription and mRNA Processing
 (A) Key transcription steps regulated by BRD4 include gene commitment, Pol II initiation, promoter-proximal pausing, and pause release. Shown are potential BRD4 contacts with acetylated H4 lysines (K5/K8/K12/K16) and H3 lysines (K9/K14) on nucleosomes, noncoding RNA (ncRNA), m⁶A-modified ncRNA reader protein, transcription factor (TF), chromatin regulator (CR), Mediator, Pol II pausing and release factors including NELF, DSIF, PAF1c, SPT6, P-TEFb, and 3'-end RNA processing and termination factors such as CPSF and CstF.
 (B) Schematic drawing showing four classes of BRD4-interacting proteins as BD binders, pNPS binders, ET binders, and CTM binders that include acetylated (Ac) histones, transcription factors (TFs), chromatin regulators (CRs), and noncoding RNA (ncRNA).
 (C) Protein-protein interaction (PPI) network showing BRD4-associated factors involved in mRNA splicing. The network is generated by STRING based on MCL (Markov Cluster

Algorithm) clustering of the proteomics datasets from immunoprecipitated BRD4-L and BRD4-S complexes.³³

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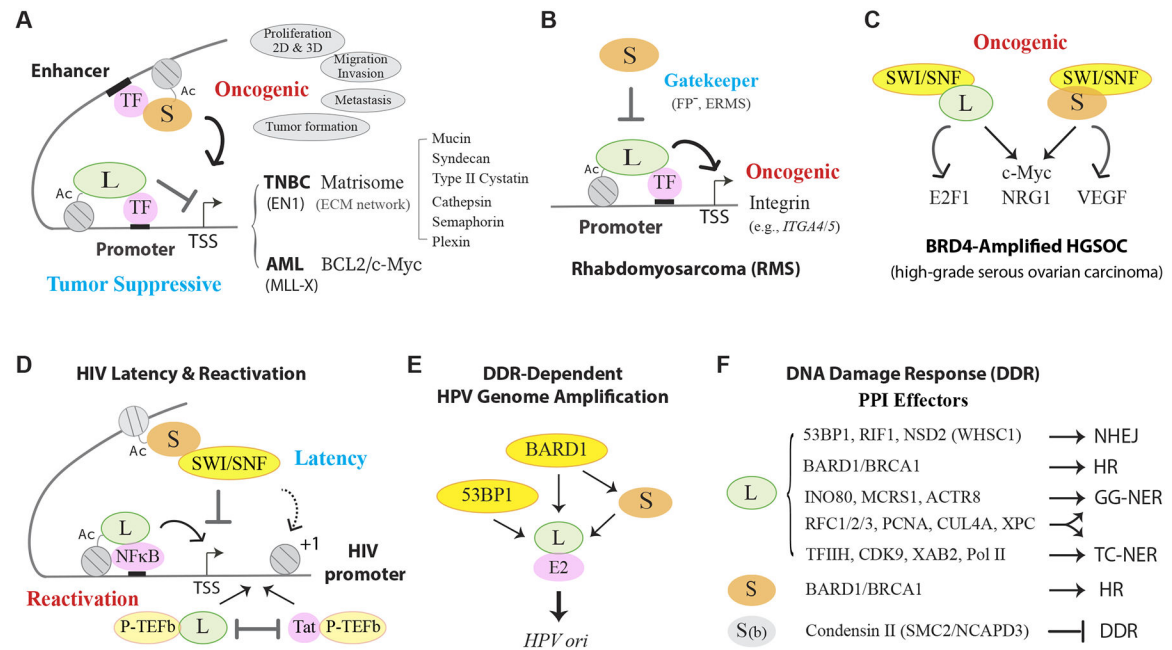


Figure 5. Distinct Functions of BRD4 Isoforms in Cancers and Virus-Infected Cells

(A) Opposing functions of BRD4 isoforms, with oncogenic BRD4-S and tumor-suppressive BRD4-L, in triple-negative breast cancer (TNBC) and mixed-lineage leukemia (MLL) gene-rearranged acute myeloid leukemia (AML).

(B) BRD4-S acts as a gatekeeper to suppress BRD4-L-activated integrin gene expression in fusion-negative embryonal rhabdomyosarcoma (FP⁻ ERMS).

(C) BRD4-S and BRD4-L recruit SWI/SNF to activate both common and distinct oncogene expression in BRD4-amplified high-grade serous ovarian cancer (HGSOc).

(D) HIV latency and reactivation regulated by distinct BRD4-L and BRD4-S isoforms.

Enhancer-associated BRD4-S recruits SWI/SNF to silence HIV gene transcription through +1 nucleosome, in contrast to BRD4-L-enhanced HIV transcription through upstream NFκB-mediated Pol II initiation and downstream P-TEFb-released Pol II pausing that can also be alleviated by HIV Tat-recruited P-TEFb. Also indicated is the competing binding to P-TEFb by mutually exclusive BRD4-L and Tat.

(E) BRD4-L recruits BRD4-S and sequential DNA damage response (DDR) factor (53BP1, BARD1) entry to HPV E2-associated origin of replication (*ori*) during HPV genome amplification in differentiated keratinocytes.

(F) Recruitment of select DDR factors by BRD4-L and BRD4-S to maintain genome integrity through non-homologous end joining (NHEJ), homologous recombination (HR), global genome nucleotide excision repair (GG-NER), and/or transcription-coupled nucleotide excision repair (TC-NER) pathways, whereas BRD4-S(b) seemingly suppresses DDR via recruitment of condensin II complexes.

(C) Schematic drawing showing other PTMs identified in BRD4, including methylation (Me), acetylation (Ac), ubiquitination (Ub), sumoylation (SM), and hydroxylation (HO). Validated PTM sites are marked by a larger bolded font, along with the identified enzymes. The 14 HO-modified lysine (K) residues identified in the BID are also indicated in parentheses. MS-identified PTMs are obtained from PhosphoSitePlus® (PSP).

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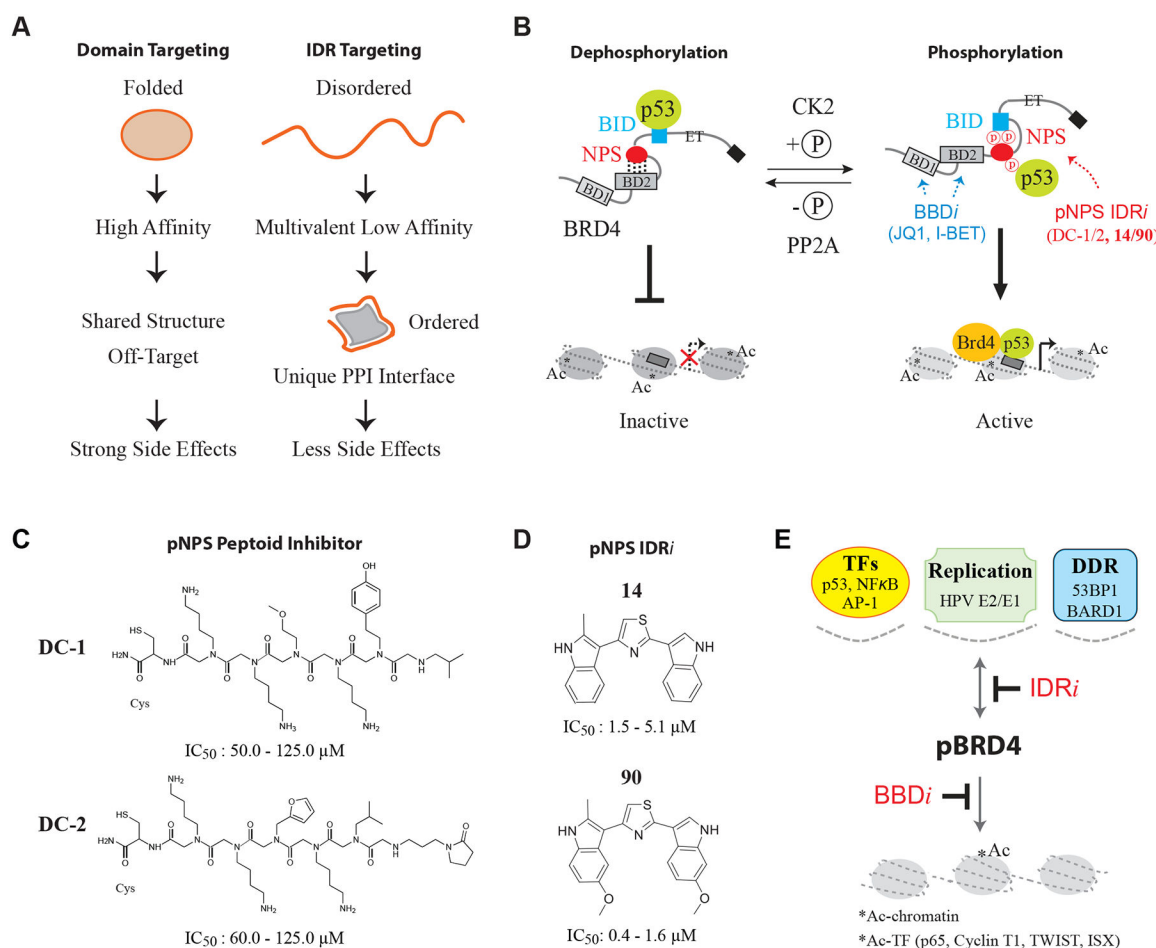


Figure 7. Phospho-BRD4 IDR-Targeting Strategy

(A) Targeting design based on folded domains or intrinsically disordered regions (IDRs).

(B) Phospho switch model for bromodomain closure/opening and factor recruitment via phosphorylation-dependent intra- and inter-molecular contact switches mediated by casein kinase II (CK2) and protein phosphatase 2A (PP2A). The regions of BRD4 targeted by BET bromodomain inhibitors (BBD*i*; e.g., JQ1 and I-BET) as well as pNPS-binding peptoids (DC-1 and DC-2) and small-molecule inhibitors (**14** and **90**) are indicated. Ac, acetylated.

(C) Chemical structure and half-maximal inhibitory concentration (IC₅₀) of pNPS peptoid inhibitors DC-1 and DC-2. The IC₅₀ range is determined by analyzing peptoid-inhibited phosphor-PDID (pPDID) interaction with HPV16 E2 (16E2) DNA-binding domain (DBD) and the p53 C-terminal regulatory domain (REG) in GST pulldown assay.²⁸

(D) Chemical structure and IC₅₀ of pNPS PPI inhibitors (IDR*i*) **14** and **90**. The IC₅₀ range is determined by analyzing compound-inhibited pPDID interaction with 16E2 DBD, dual BRCT domains of 53BP1 and BARD1, and p53 REG using AlphaScreen.³³

(E) Mode of compound action. IDR*i* inhibit phospho-BRD4 (pBRD4) interaction with select TFs, replication factors, and DDR factors without globally altering pBRD4 chromatin-binding activity, whereas BBD*i* work primarily by blocking pBRD4 binding to acetylated (Ac) chromatin and/or Ac-TFs.